Andrology & Sexual Medicine

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 66	Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System
Ch 67	Male Reproductive Physiology
Ch 68	Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome
Ch 69	Male Infertility
Ch 70	Surgical Management of Male Infertility
Ch 71	Physiology of Penile Erection and Pathophysiology of ED
Ch 72	Evaluation and Management of Erectile Dysfunction
Ch 73	Priapism
Ch 74	Disorders of Male Orgasm and Ejaculation
Ch 75	Surgery for Erectile Dysfunction
Ch 76	Diagnosis and Management of Peyronie Disease
★	Putting it together: andrology decisions

66 Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

1 · Scrotum

Each layer is a surgical plane carried down from the abdominal wall.

Layer (superficial to deep)	Continuous with	Note
Tunica dartos	Scarpa, Colles, penile dartos	Smooth muscle
External spermatic fascia	External oblique	—
Cremasteric fascia	**Internal oblique**	**Striated** cremaster; thermoregulation
Internal spermatic fascia	Transversalis fascia	—
Tunica vaginalis (parietal, visceral)	Peritoneum	Hydrocele collects between its layers; visceral layer fused to albuginea

- Anterior wall: external pudendal vessels, ilioinguinal and genitofemoral nerves. Posterior wall: perineal vessels and nerve. Nothing crosses the raphe.
- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *scrotum to ipsilateral superficial inguinal; testis to para-aortic/interaortocaval; cauda to external iliac*

Not in the paper answer:

- Vessels and nerves run parallel to the rugae; the posterior femoral cutaneous nerve also supplies the posterior wall.

> **KEY POINT**
> Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum. †So scrotal violation in testis cancer adds the inguinal nodes to the at-risk field.

2 · Testis

Vessel anatomy decides biopsy site and safe varicocele surgery.

Structure

- Volume is **15–25 mL**, length 4.5–5.1 cm. The capsule is visceral tunica vaginalis, **tunica albuginea** (with smooth muscle) and tunica vasculosa.
- The **mediastinum testis** is the posterior echogenic band on ultrasound (US). The appendix testis is a **müllerian** remnant.
- About 250 lobules hold **600–1200 seminiferous tubules**, which drain to the rete testis.
- The interstitium is **20–30%** of volume and holds the **Leydig cells**. **Sertoli** tight junctions form the blood–testis barrier.

Arterial supply — three sources

Artery	Origin	Point
Testicular (internal spermatic)	Aorta below renal artery	Main supply; larger than the other two combined
Deferential	Superior (or inferior) vesical	Vas, cauda; key collateral, lost after vasectomy

Artery	Origin	Point
Cremasteric	Inferior epigastric	Tunica vaginalis; anastomoses at mediastinum

- Collaterals may save the testis after testicular artery ligation, but atrophy or azoospermia can occur.

> **KEY POINT**
> **Surgical rules**: no traction suture through the lower pole. Biopsy the **medial or lateral upper pole**; the midportion has fewest vessels. After vasectomy the deferential collateral may be gone, so preserve the testicular artery in later cord surgery and counsel about devascularisation.

Veins — pampiniform plexus

The plexus surrounds the artery and cools it by **countercurrent heat exchange** to **2–4 °C below rectal** temperature. A varicocele disrupts this.

Group	Accompanies	Drains to
Anterior	Testicular artery	Inferior vena cava (IVC) on the right; left renal vein on the left
Middle	Vas	Pelvic veins
Posterior	Posterior cord	Epigastric and pudendal veins

- P1 Varicocele – Pathophysiology — *left vein into renal vein (**~90% of varicoceles left**), oblique right entry valve-like; middle/posterior groups cause recurrence*

Not in the paper answer:

- **High (retroperitoneal) ligation** takes artery and vein above the deferential and external vessels; collaterals suffice.
- **Inguinal or subinguinal** ligation takes veins only, because arterial ligation there risks devascularisation more.
- Pudendal, cremasteric and vasal anastomoses explain recurrence after embolisation.

Lymphatics and nerves

- Dividing lymphatics at varicocelectomy causes a **hydrocele**; microsurgery preserves them.
- Supply is autonomic (renal and aortic plexuses, and pelvic plexus via the vas), with **no somatic innervation**. The genital branch of the genitofemoral nerve supplies the tunica vaginalis.
- The **"trifecta" nerve complex** (perivasal, posterior periarterial/lipomatous, intracremasteric) is the target of microsurgical cord denervation for chronic orchialgia.

Blood–testis barrier

Sertoli–Sertoli tight junctions divide basal from adluminal compartments; capillary endothelium and peritubular myoid cells add to the barrier *(see Ch 67)*.

Imaging

Modality	Role
Ultrasonography (US), 7.5–10 MHz with Doppler	**First line**: pain, mass, swelling, epididymo-orchitis, recurrent varicocele
Computed tomography (CT)	Not primary; cord or inguinal disease; air in Fournier gangrene
Magnetic resonance imaging (MRI)	Second line: indeterminate US, intra- versus extratesticular; nonpalpable testis (MR angiography)

- If torsion is suspected, **explore**; Doppler must not delay surgery.
- Varicocele is a **clinical diagnosis**. Use US (standing, Valsalva) only if examination is equivocal.

3 · Epididymis

The epididymis matures and stores sperm.

- Caput, corpus and cauda lie on the posterolateral testis. **Principal** cells form the blood–epididymis barrier; there are also basal cells. **Clear** cells acidify the lumen, which keeps stored sperm quiescent. Cilia occur only in the efferent ducts.
- Testicular artery supplies caput and corpus, deferential the cauda; cremasteric contributes.
- Nerves and smooth muscle increase toward the cauda, which shifts from storage to propulsion.
- A minority have an inverted epididymis, with the caput at the lower pole.

Wolffian (mesonephric) duct	Müllerian (paramesonephric) duct
Epididymis, vas, seminal vesicle, ejaculatory duct, **appendix epididymis** (efferent ductules from mesonephric tubules)	**Appendix testis**, prostatic utricle

4 · Vas deferens

The vas is the target of vasectomy and reconstruction.

- From the cauda it is convoluted for 2–3 cm, then runs straight, **posterior** to the cord vessels, through the inguinal canal to the pelvic side wall. It crosses **anterior to the ureter**, forms the ampulla and joins the seminal vesicle.
- It is **30–35 cm** long with a **0.2–0.7 mm** lumen and has **three muscle layers** (longitudinal, circular, longitudinal). Lymph drains to external and internal iliac nodes.
- About half the cord nerves surround the vas: sympathetic fibres drive emission, sensory fibres cause chronic scrotal pain.
- **Vasogram**: through a hemivasotomy, a 24 G angiocatheter injects indigo carmine or contrast. It can cause obstruction, so do it **only at reconstruction**.
- US confirms congenital vasal absence if examination is equivocal.

5 · Seminal vesicles and ejaculatory ducts

These glands make most of the semen.

- Behind the bladder base, they join the ampulla as ejaculatory ducts opening on the verumontanum at **5 and 7 o'clock**.
- They bud from the mesonephric duct at 12 weeks, linked to the ureteric bud, so **renal anomalies coexist**.
- Arteries: vesiculodeferential and inferior vesical; lymph to internal iliac nodes. Noradrenergic nerves contract them at emission.
- The fluid is ~70% of semen and alkaline. It carries **fructose**, prostaglandins and **semenogelin**, which forms the coagulum later liquefied by prostate-specific antigen (PSA).
- On **transrectal ultrasound (TRUS)**, a post-ejaculation anteroposterior (AP) diameter **>16 mm** or volume **>6 mL** means obstruction. TRUS also guides aspiration and vesiculography.
- **MRI** is the modality of choice; CT is for stones. Vesiculoscopy uses 4.5–6.9 Fr scopes.

KEY POINT
A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction.

6 · Prostate

Zones explain where cancer and benign enlargement arise.

- P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — *McNeal zones (peripheral **70%**, transition gives BPH), fasciae, arteries, DVC, lymph, cavernous nerves*

Not in the paper answer:

- PSA normally stays in the lumen; disrupted architecture lets it reach serum. Cancer means **loss of the basal cell layer**.
- The "capsule" is 0.5–2 mm thick and deficient at the apex.
- Parasympathetic nerves drive secretion; sympathetic (**α1**) nerves drive expulsion at emission.
- TRUS volume is length × width × height × **0.52**, which underestimates.
- **Multiparametric MRI** (T2, diffusion, dynamic contrast) is scored by Prostate Imaging Reporting and Data System (PI-RADS) v2.1; targeted biopsy finds less insignificant cancer.
- Extraprostatic arterial anastomoses matter for embolisation.
- The normal gland is **~25 mL**; the transition zone grows fastest with age. The peripheral zone is hyperechoic on TRUS.
- The internal (preprostatic) sphincter keeps continence at rest.

7 · Urethra

Segment anatomy decides how a stricture is imaged and treated.

- P1 Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias — *segments (membranous **2–2.5 cm**; Cowper 3/9 o'clock), epithelium, sphincter (Onuf), blood supply*

Not in the paper answer:

- Sonourethrography images anterior strictures; retrograde urethrogram (RUG) shows stricture and trauma (membranous narrowing is normal); MRI shows malignancy.

8 · Penis

Erection depends on the tunica, cavernosal arteries and veno-occlusion.

Structure

- There are paired **corpora cavernosa** (crura on the pubic arch) and a **corpus spongiosum** (bulb, urethra, glans).
- The septum is incomplete distally, so the corpora act as **one unit** (†one-sided injection fills both).
- The **tunica albuginea** (2–3 mm) has an inner **circular** layer forming intracavernosal struts and an outer **longitudinal** layer from glans to inferior pubic ramus; it is thin ventrally.
- **Buck fascia** encloses the corpora and splits around the spongiosum. From midline outward it holds the **deep dorsal vein, then arteries, then nerves**.
- Sinusoidal smooth muscle is **~45%** of volume. Endothelial nitric oxide synthase (NOS) sustains erection.

Arteries

The external pudendal artery supplies skin. The internal pudendal artery runs through Alcock canal to become the **common penile artery**:

Branch	Supplies
Bulbourethral	Spongiosum, urethra, glans
Cavernosal	Enters at hilum, runs centrally; helicine arteries fill the sinusoids (erection)
Dorsal	Under Buck fascia to the glans; circumflex branches

Accessory pudendal arteries may supplement or replace.

Veins, nerves

- The superficial dorsal vein (outside Buck fascia) drains skin; the **deep dorsal vein** (under it) drains to the periprostatic plexus.
- In the distal two-thirds, sinusoids drain via the subtunical plexus and **emissary veins** to circumflex veins and the deep dorsal vein. The proximal third drains via cavernous and crural veins to the internal pudendal.
- Shaft skin is mobile; glans skin is fixed.
- The **dorsal nerve** (pudendal, **S2–S4**) gives sensation; the **cavernous nerves** (pelvic plexus) give erection.

> **KEY POINT**
> Erection is veno-occlusive: filled sinusoids compress subtunical and emissary veins against the inelastic tunica. †Tunical or smooth-muscle damage defeats this and causes venous leak.

Penile duplex (after intracavernosal injection)

Parameter	Threshold	Meaning
Peak systolic velocity	**<30 cm/s**	Arterial insufficiency
End-diastolic velocity	**>5 cm/s**	Veno-occlusive dysfunction
Resistive index	**<0.80**	Veno-occlusive dysfunction

MRI shows fracture, prosthesis malposition, tumour.

67 Male Reproductive Physiology

1 · Hypothalamic–pituitary–gonadal axis

The hypothalamic–pituitary–gonadal (HPG) axis governs sex development, puberty, testosterone and sperm; every hormonal work-up reads it. Pulse frequency and feedback set target responsiveness.

	Peptide hormones	Steroid hormones
Examples	Luteinising hormone (LH), follicle-stimulating hormone (FSH)	Testosterone, oestradiol
Storage	Secretory granules	Not stored; secretion equals production
Receptor	Cell surface, acting via cyclic AMP (cAMP)	Intracellular, altering transcription; carried bound in plasma

Hypothalamus

- Gonadotrophin-releasing hormone (GnRH) is a decapeptide from the preoptic and arcuate nuclei. Via the portal system, its only function is LH/FSH release. Half-life is **5–7 min**.
- Its rhythms are pulsatile (every **90–120 min**), circadian (testosterone peaks early morning) and seasonal.
- **Pulsatility is essential**: a continuous GnRH agonist (leuprolide) desensitises the pituitary, giving castrate testosterone.
- In **Kallmann syndrome** GnRH neurones fail to migrate, which causes hypogonadotrophic hypogonadism. †Anosmia coexists because the neurones migrate with the olfactory placode.

GnRH modulation	Factors
Negative	Opioids (β-endorphin), FSH, LH, testosterone
Positive	Prostaglandin E2, insulin, **kisspeptin** (puberty), **leptin**
Variable	Dopamine (catecholamines)

Anterior pituitary

- LH and FSH are glycoproteins with a common **α subunit** (shared with thyroid-stimulating hormone [TSH] and human chorionic gonadotrophin [hCG]) and a specific **β subunit**. Both act via cAMP.
- **Prolactin** may raise Leydig LH receptors and support libido. **Hyperprolactinaemia abolishes gonadotrophin pulsatility**, so testosterone falls. Low prolactin is not pathological.

	LH	FSH
Pulses	8–16/24 h; follow GnRH	Smaller; partly GnRH-independent (inhibin, activin)
Target	**Leydig cell**	**Sertoli cell** (and spermatogonia)
Action	Steroidogenesis: cholesterol to pregnenolone to testosterone	Tubule growth; **initiates spermatogenesis at puberty**; maintains quantitative output
Main feedback	Testosterone (hypothalamus)	Oestradiol (pituitary) and **inhibin B**

Testicular hormones and feedback

- Testosterone output is ~**5 mg/day**, pulsatile. 5α-reductase converts it to **dihydrotestosterone (DHT)**, which is more potent and needed in most peripheral tissues (not testis or muscle). Aromatase converts it to **oestradiol**, which feeds back especially on FSH.
- FSH makes Sertoli cells produce androgen-binding protein (ABP), transferrin, lactate, inhibin and growth factors.
- **Inhibin** (Sertoli, 32 kDa) suppresses FSH. **Activin**, of the transforming growth factor-β (TGF-β) family, stimulates FSH.

> **KEY POINT**
> There are two loops: LH with Leydig cells (testosterone), and FSH with Sertoli cells (inhibin, oestradiol). So **LH reports Leydig function, while FSH and inhibin B report the tubules**. High FSH with low inhibin B means spermatogenic failure.

Development and ageing

- **SRY** (Yp) encodes an HMG-box protein that bends DNA, starting the testis. **DAX1** and **WNT4** are "antitestis" genes that can induce female development despite SRY; the female pathway is not a passive default.

Fetal cell	Product	Effect
Leydig	Testosterone	Wolffian duct forms internal genitalia
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testicular descent
Target tissue	DHT	External genitalia
Sertoli	Müllerian-inhibiting substance (MIS/AMH)	Müllerian regression; keeps germ cells out of meiosis

- A neonatal surge precedes childhood quiescence; puberty (~12 years) is triggered by kisspeptin, leptin and melatonin.
- **Ageing**: testosterone is ~one-third lower by the 7th decade and FSH ~threefold higher. Leydig cells fall, sex hormone-binding globulin (SHBG) rises, diurnal rhythm is lost and feedback is blunted, so failure is both peripheral and central.

2 · Testis

Leydig cells make testosterone; Sertoli cells nurse sperm.

Architecture and blood flow

- It has **600–1200 tubules** and ~**700 million Leydig cells** in young adults.
- The cord holds one artery in ~half, two in ~30% and three in ~20%; at microsurgical varicocelectomy, ligating a single artery can cause atrophy.
- **Countercurrent exchange** cools arterial blood **2–4 °C** below core and recycles testosterone from vein to artery. It is lost in varicocele and cryptorchidism.
- Rete and efferent ductules reabsorb tubular fluid under oestrogen control.

Leydig cells and testosterone synthesis

- They differentiate by week 7 under hCG/LH; a neonatal wave at ~2–3 months imprints hypothalamus, liver and prostate.

- **Rate-limiting step**: LH raises cAMP, and **steroidogenic acute regulatory (StAR) protein** moves cholesterol into the inner mitochondrial membrane to become pregnenolone.
- Smooth endoplasmic reticulum enzymes (3β-HSD, 17α-hydroxylase/17,20-lyase, 17β-HSD) then make testosterone. Mutations cause 46,XY disorders of androgen biosynthesis.
- FSH and prolactin enhance LH response; oestrogens and androgens inhibit.
- †In blood, ~44% of testosterone is bound tightly to SHBG, ~54% loosely to albumin and ~2% is free. Free plus albumin-bound is "bioavailable".

Testosterone peak	Timing	Role
Fetal	12–18 weeks	Builds reproductive tract
Neonatal	~2 months	Imprints target tissues
Pubertal	12–17 years	Masculinisation
Adult	Peak in 2nd–3rd decade, then decline	Maintains androgen-dependent organs

Sertoli cells and blood-testis barrier

- P3 P1 Spermatogenesis; Male reproductive immunology — *Sertoli nurse functions, ABP (intratubular androgen ~**50-fold** serum), basal/adluminal compartments, anatomical + functional barrier, no antibodies before puberty, chemotherapy sanctuary*

Not in the paper answer:

- Sertoli tight junctions are the body's strongest, yet open to let spermatocytes pass.
- **Inhibin B** marks Sertoli function. Sertoli cells also signal via c-kit/kit ligand and perform spermiation.
- Peritubular myoid cells are contractile and secrete P-Mod-S, which modulates Sertoli function.
- TGF-β3 and tumour necrosis factor-α (TNF-α) open the barrier reversibly by reducing occludin and ZO-1.

3 · Spermatogenesis

A semen test reflects events about two months earlier.

- P3 Spermatogenesis — ***~64 days*** *(42–76); proliferation (Ad, Ap every 16 days, B), then meiosis, then spermiogenesis (acrosome from Golgi, flagellum from centriole), then spermiation; six stages, helical wave; cytoplasmic bridges*

Not in the paper answer:

- Output is ~**123 million/day**, about 1000 per heartbeat.
- Primary spermatocytes pass through preleptotene, leptotene, zygotene and pachytene. Spermatids mature through Sa, Sb, Sc, Sd1 and Sd2.
- A **cycle** is the succession of stages at one point; a **stage** is one cell association; a **wave** is the arrangement along the tubule.
- †Spermatogenesis needs high intratesticular testosterone, so exogenous testosterone suppresses it.

Stem cells

- Primordial germ cells migrate to the gonadal ridge as gonocytes, then spermatogonia; failed migration causes extragonadal germ cell tumours or Sertoli cell–only testes.
- Stem cells self-renew through kit ligand/c-kit and glial cell line–derived neurotrophic factor (GDNF).
- "Selfish spermatogonia" (FGFR2/FGFR3 mutations) cause paternal-age disorders and spermatocytic seminoma.

Meiosis

	Mitosis	Meiosis
Divisions / daughters	1 / 2 identical	2 / 4 varied, haploid
Synapsis, crossover	None	Prophase I (synaptonemal complex, chiasmata)
Centromere division	Anaphase	**Anaphase II**

- **Defective recombination** occurs in ~half of maturation arrest. Altered crossover causes aneuploidy, so offspring of in vitro fertilisation and intracytoplasmic sperm injection (IVF/ICSI) have more chromosomal abnormalities.

Genetics

Y microdeletions of the azoospermia factor (AZF) region occur in **15%** of azoospermic and 6–8% of severely oligospermic men, the commonest defined molecular cause.

Deletion	Histology	Sperm retrieval
Complete AZFa	Sertoli cell–only	Highly unlikely
Complete AZFb	Maturation arrest	Highly unlikely
AZFc	Hypospermatogenesis / focal spermatogenesis	**Often found**
Partial AZFa/b	—	Ejaculated sperm possible

X-linked genes: SOX3, TEX11. **Paternal age** matters because spermatogonia divide ~23 times a year, raising de novo dominant mutations (achondroplasia, Apert, Marfan), autism, schizophrenia, aneuploidy and **DNA fragmentation**, with poor embryos and miscarriage.

4 · Epididymis

The epididymis makes sperm motile and fertile.

- The 3–4 m duct runs from caput (8–12 efferent ductules) through corpus to cauda. Testicular sperm are **immotile and cannot fertilise**, except by ICSI.
- The epithelium has principal cells ("stereocilia" are microvilli) and basal cells; ciliated cells occur only in efferent ductules. Its barrier is less tight than the testis. Muscle and sympathetic nerves increase toward the cauda.

Functions

- **Transport** takes **2–12 days**, driven mainly by spontaneous rhythmic contraction. Recent ejaculation shortens only cauda transit.
- **Storage**: the cauda holds about half of 155–209 million sperm per side; prolonged storage increases DNA fragmentation.
- **Maturation** gives progressive motility and zona binding and penetration, through surface charge change, disulfide cross-linking and membrane remodelling. **Fertility is reached mostly by the late corpus or early cauda**.
- Epididymosomes deliver small RNAs that sperm carry to the embryo.

> **KEY POINT**
> Pregnancy occurs even after vasoepididymostomy at the efferent ductules, but **the more distal the anastomosis, the better the fertility**. After vasectomy for >15 years, motility is lower after reversal because of epididymal dysfunction.

- The epididymis is **androgen dependent**, needing higher levels mainly as DHT, which explains impaired semen on 5α-reductase inhibitors.
- **Heat** abolishes storage, and **denervation** (spinal cord injury, diabetes) impairs transport.

5 · Vas deferens, seminal vesicles, ejaculatory ducts

Semen chemistry shows whether these organs are present and open.

- The vas is wolffian and 30–35 cm long. With a **muscle:lumen ratio of ~10:1** it is the most muscular hollow viscus. Its lumen is 200–700 µm.
- Adrenergic (hypogastric) and purinergic nerves drive peristalsis before emission; the testicular-side segment has fewer nerves. Vasal reserve is ~130 million sperm.
- The **seminal vesicle** is a 5–7 cm coiled tubule that gives ~**80%** of seminal volume in the late fractions. Its fluid is alkaline and carries **fructose** and prostaglandins; **semenogelin I** forms the coagulum. Semen antioxidants include glutathione peroxidase, superoxide dismutase (SOD) and catalase.
- The **ejaculatory duct** has no sphincter; reflux is prevented by its **acute insertion angle**.
- Vesicle and duct behave like bladder and urethra: obstruction may be **physical** or **functional** (vesicle myopathy from diabetes or drugs). Static transrectal ultrasound (TRUS) may not tell them apart.
- An **ejaculate with pH <7.2, low volume and no fructose** means absent or blocked seminal vesicles, as in congenital bilateral absence of the vas (CBAVD) or ejaculatory duct obstruction.

6 · Spermatozoon, capacitation and fertilisation

Structure explains motility defects and some ICSI failures.

Part	Contents
Head (4.5 × 3 µm)	Compacted nucleus; acrosome (enzymes)
Neck	Proximal centriole; axoneme starts
Midpiece	~75 mitochondria in a helix, outer dense fibres; ends at annulus
Principal piece / endpiece	Axoneme with fibrous sheath

- Length is ~60 µm. The axoneme is **"9 + 2"**, and **dynein** (a Mg^{2+}-ATPase) slides the doublets. Inner dynein arm mutants have no motility; outer arm mutants have reduced motility.
- Sperm mitochondrial DNA (mtDNA) mutates 40–100× faster, but **paternal mitochondria are degraded** in the egg (by ubiquitin), so inheritance is maternal.
- **Capacitation** in the female tract gives hyperactivated motility and readiness for the **acrosome reaction**. The coagulum prevents premature capacitation.
- Binding to zona **ZP3** triggers the acrosome reaction. **Fertilin (PH30)** mediates fusion with the oolemma. Sperm **phospholipase C zeta (PLCζ)** triggers calcium oscillations, which activate the oocyte.

Sperm defects

- Morphology is graded by strict (Kruger) criteria. Head defects rarely reflect DNA quality, except **pinhead** and **multiple-tailed** sperm.
- In **globozoospermia** (round head, no acrosome) fertilisation fails even with ICSI, because of PLCζ deficiency.

	Ciliary dyskinesia (immotile cilia)	MMAF (multiple morphological abnormalities of the flagella; fibrous sheath dysplasia)
Motility	<10%	Severely reduced
Morphology	Normal on light microscopy; axonemal defect on electron microscopy (EM)	Short, bent, coiled or absent flagella
Cause	Recessive; respiratory disease, dextrocardia (Kartagener)	Genetic; also varicocele, reactive oxygen species (ROS)
Treatment	IVF-ICSI	Correct cause; else IVF-ICSI

7 · Ejaculation physiology †

Each cause of ejaculatory failure hits a specific phase.

ALGORITHM

1. **Emission** (sympathetic, T10–L2 via the hypogastric nerve; α1): vas, seminal vesicle and prostate contract and deliver semen to the posterior urethra.
2. The **bladder neck closes**. If it fails to close, then ejaculation is retrograde.
3. **Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus and pelvic floor contractions, with a relaxed external sphincter, give an antegrade ejaculate.

- P3 Retrograde ejaculation — *failed bladder neck closure (α-blocker, diabetes, bladder neck surgery, RPLND) gives retrograde; loss of emission gives anejaculation*

Not in the paper answer:

- Fraction order: sperm-rich and prostatic first, seminal vesicle last.

68 Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

1 · The men's health centre

The urologist is often the only doctor a man sees.

- The centre is multidisciplinary (urology, primary care, cardiology, gastroenterology). Men avoid preventive care but seek help "below the belt" (erectile dysfunction [ED], infertility, voiding), so the urologist is the **front door** and triages the whole man.
- Advanced practice providers and telemedicine extend reach; decisions are shared.

Age band	Dominant need	Key partners
18–39	Sexual, reproductive, testicular ("rule out hypogonadism"); preventive	Sports medicine, physiotherapy, mental health
40–64	Prostate and colon screening; disease management	Gastroenterology (GI), cardiology, endocrinology, bariatrics, pulmonology
65+	Falls, neurodegeneration, hearing/vision, nutrition	Geriatrics, neurology, audiology, ophthalmology

2 · The longevity gap

Men die earlier.

- Male life expectancy was **76.1** versus 81.1 years for females (2016).
- Males have higher mortality in **12 of 15** leading causes; the largest gap is **suicide (>3-fold)**. Men have more early myocardial infarction (MI), and type 2 diabetes (T2DM) is rising faster in them.
- Causes: hormonal (testosterone harm, oestrogen cardioprotection), behavioural, risk-taking.
- Prostate cancer is the only US malignancy with rising stage IV at diagnosis, so discuss screening.

3 · Urologic screening

Questionnaires catch unvolunteered problems.

- Screen sexual and voiding complaints together; check prostate-specific antigen (PSA) and urinalysis per guidelines.

Questionnaire	Screens
Sexual Health Inventory for Men (SHIM)	Erectile function
Androgen Deficiency in the Aging Male (ADAM)	Androgen-deficiency symptoms
International Prostate Symptom Score (IPSS)	Lower urinary tract symptoms (LUTS)

4 · Urologic disorders of focus

These are often the first signs of systemic disease.

Testosterone deficiency (TD)

- P4 Male menopause (andropause, late-onset hypogonadism) — *low T on more than one occasion + symptoms; AUA **<300 ng/dL** on 2 morning samples; LH separates primary/secondary, prolactin if LH low; TRT contraindications; monitor T, haematocrit (**>54%** hold), PSA at 3–6 months then yearly*

Not in the paper answer:

- For haematocrit, the American Urological Association (AUA) withholds testosterone replacement therapy (TRT) at >50% and the Endocrine Society at >54%; then reduce dose or phlebotomise. A PSA rise or abnormal digital rectal examination needs urology review.
- Chronic opioids (~5 million US men): the Endocrine Society advises screening and considering TRT.
- **Randomised trials show TRT benefit** for obesity, metabolic syndrome, diabetes, muscle and bone. Mood, cognition and sexual benefits are only observational.
- Testosterone (T) and depression share causes and symptoms, but **TRT is not a treatment for depression**. The likeliest responders are clearly deficient, <60 years old, with milder depression.
- T falls with age from testicular and hypothalamic–pituitary defects; symptoms overlap with comorbidity. Exclude secondary causes before TRT.

> **KEY POINT**
> † **Fertility caution**: exogenous T suppresses LH/FSH and intratesticular T, which causes oligo- or azoospermia. Men wanting children should avoid TRT and use clomiphene, human chorionic gonadotrophin (hCG) or aromatase inhibitors *(see Ch 69)*.

† TRT	Points
Contraindications	Near-term fertility wish; breast cancer; untreated prostate cancer; haematocrit >50%; untreated severe obstructive sleep apnoea (OSA); uncontrolled heart failure; MI/stroke within 3–6 months
Target	Mid-normal T (≈450–600 ng/dL)
Formulations	Gels, injections, pellets, each with its own monitoring plan

Erectile dysfunction

- ED affects >50% of men aged 40–70; causes are multiple.
- **Every man with ED, of any age, is at cardiovascular disease (CVD) risk**. †Penile arteries are small, so ED precedes coronary events by ~2–5 years.
- Treatment steps from lifestyle to phosphodiesterase type 5 (PDE5) inhibitors (never with nitrates), then intracavernosal injection, intraurethral or vacuum therapy, then prosthesis (highest satisfaction; counsel irreversibility, infection, malfunction).
- Platelet-rich plasma and low-intensity shock wave: trials only.

BPH and infertility

- Benign prostatic hyperplasia (BPH) is mostly treated medically; refer when medication fails, is not tolerated or is not wanted.
- Infertility is no pregnancy after 1 year; a male factor is present in ~half; semen analysis is the cornerstone *(see Ch 69)*.

5 · Cardiovascular prevention

Statins and blood pressure control are the main levers.

- The American College of Cardiology/American Heart Association (ACC/AHA) ASCVD calculator gives 10-year risk but ignores family history and diabetes control.
- Only **lowering low-density lipoprotein cholesterol (LDL-C)** benefits primary prevention; raising high-density lipoprotein (HDL) or lowering triglycerides does not. Screen lipids from age 20.

Statin decision (ACC/AHA)	Action
LDL-C ≥190 mg/dL	High-intensity statin with or without a nonstatin
LDL-C 70–189, 10-year risk ≥ **7.5%**	Moderate–high intensity statin
Risk 5–7.5%	Moderate intensity may be started
Statin-intolerant, risk >20%	Ezetimibe or PCSK9 inhibitor

- Primary prevention prefers moderate intensity. USPSTF, NICE and ESC start at ≥10% risk.
- A **coronary artery calcium (CAC) score** suits asymptomatic men >40, most usefully at **intermediate risk**; zero means <1% atherosclerosis (less reliable in diabetics).
- Lipoprotein(a) risk is near-linear; statins do not lower it.

Blood pressure (BP) (ACC/AHA 2017)	Systolic	Diastolic
Normal	<120	and <80
Elevated	120–129	and <80
Stage 1	130–139	or 80–89
Stage 2	≥140	or ≥90

Hypertension is the most important modifiable risk. Elevated or stage 1 BP gets lifestyle change (salt, weight, DASH diet, exercise, alcohol). Drugs: stage 2, or stage 1 with age ≥65, T2DM, CVD, chronic kidney disease or 10-year risk ≥10%.

6 · Obesity and metabolic syndrome

Fat lowers testosterone, and low testosterone adds fat.

- Body mass index (BMI): overweight 25–29.9; obesity I 30–34.9, II 35–39.9, III ≥40.
- Gaining ≥5 kg after 20 raises coronary and T2DM risk regardless of BMI.
- Advise weight loss at BMI ≥30, or 25–29.9 with CVD risk, OSA or osteoarthritis; GLP-1 agonists (semaglutide) are FDA-approved.
- †Metabolic syndrome (ATP III) needs ≥3 of: waist >102 cm, triglycerides ≥150 mg/dL, HDL <40 mg/dL, BP ≥130/85, glucose ≥100 mg/dL. Low T and metabolic syndrome drive each other, because adipose aromatase raises oestradiol, which lowers LH.

7 · GI, skin and mental health

These often present to the urologist.

- Evaluate **constipation** in every man with urinary tract infection (UTI) or retention. Alarm features need colonoscopy.
- Iron deficiency anaemia carries >30-fold GI malignancy risk. Colorectal screening: **45–75** years. Long-term proton pump inhibitor use: check for Barrett.
- Basal cell is the commonest skin cancer; melanoma by ABCDE. Psoriasis is linked to MI. Alopecia: 5α-reductase inhibitors and minoxidil.

- Suicide is much higher in men; substance use, autism and attention-deficit/hyperactivity disorder (ADHD) are male-predominant.

69 Male Infertility

1 · Definition and overview

Half of infertility involves the man, so he must always be examined.

- Infertility is no pregnancy after **1 year** of unprotected intercourse. It affects ~15% of couples; male and female factors contribute equally.
- Most conceptions occur in the first 3–6 cycles, so evaluation may start after 6 months.
- Reproduction is **probabilistic**: a test modifies a probability; it is not a verdict.
- Idiopathic infertility is the largest group. **Varicocele is the commonest identifiable cause**, then obstruction; others are uncommon.

† Classification	Examples
Pre-testicular (endocrine)	Hypogonadotropic hypogonadism (HH), Kallmann, prolactinoma, exogenous androgens, opioids, obesity, congenital adrenal hyperplasia (CAH)
Testicular	Klinefelter, Y microdeletion, cryptorchidism, torsion, orchitis, varicocele, chemotherapy/radiation, Leydig failure
Post-testicular	Obstruction (epididymal, vasal, ejaculatory duct), congenital bilateral absence of the vas (CBAVD), ejaculatory dysfunction, antisperm antibodies, structural sperm defects

2 · History

History finds reversible causes: drugs, toxins, timing.

- P2 Discuss the role of urologist in reproductive endocrinology and male infertility practice — *history, examination, endocrine tests (LH localises, FSH/inhibin B = germ-cell mass, prolactin), HH/hyperprolactinaemia/CAH treatment, surgical roles, ART choice, testis cancer in azoospermia*

Not in the paper answer:

- Paternal age ≥40: counsel about offspring bipolar disorder, schizophrenia, autism and attention-deficit/hyperactivity disorder (ADHD) (ASRM/AUA 2021). Prior paternity excludes many congenital causes.
- Lubricants (K-Y, Astroglide, Replens, saliva) impair sperm; **Pre-Seed** is safe.
- Mnemonic **TICS**: Toxins, Infection, Childhood history, Sexual history.
- Cancer impairs spermatogenesis even before chemotherapy.
- The **female partner's age is the most important determinant**; fecundity falls after 35.
- Semen quality peaks after **1–2 days' abstinence**, so advise daily intercourse around ovulation.
- Hernia mesh can entrap the vas.

Gonadotoxins and drugs

> **KEY POINT**
> **Exogenous testosterone (T) is a contraceptive**. It is aromatised to oestradiol (E2), which feeds back on the pituitary and lowers luteinising hormone (LH). Intratesticular T then falls, and spermatogenesis falls with it. Recovery usually takes ≥4 months after stopping (after ≤2 years' use); after longer use it is unknown.

- Other harmful drugs: antiandrogens, spironolactone, corticosteroids, oestrogens, anabolic steroids, some antiretrovirals, and 5α-reductase inhibitors (5ARIs; limited effect).

- Cannabis lowers T and count dose-dependently. Alcohol increases aromatisation. Smoking causes oxidative stress and DNA damage, additive with varicocele. Maternal smoking gives sons smaller testes.

Infection and childhood conditions

- Infection injures either **anatomically** (duct scarring, as in TB) or **functionally** (leucocytes, reactive oxygen species [ROS], cytokines).
- After torsion, contralateral biopsy is usually abnormal, ~half have later spermatogenic effects, and antisperm antibodies form.

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Paternity	Near normal	Markedly reduced
Inhibin B	~⅔ of controls	~⅓ of controls

- **Laterality is the key prognostic factor.** The lesion affects germ and Sertoli cells, begins in fetal life and worsens with time.
- The American Urological Association (AUA) advises orchidopexy within **18 months**, because germ cells are lost from 15–18 months. If this was missed, surgery before 10 years still improves later sperm retrieval.

3 · Physical examination

Testis size and the vasa sort most azoospermic men.

- Eunuchoid means arm span >5 cm over height. Klinefelter habitus is often absent. Examine the man standing.
- An engorged epididymis suggests obstruction. Testis size reflects spermatogenic mass.

Testis size	Abnormal
Seager caliper (long axis)	**≤4.6 cm**
Prader orchidometer (volume)	≤20 mL
Scrotal ultrasound (US)	Most accurate; added value unclear

- If the presumed **vas** slips from the fingers three times, it is absent (Meacham). Unilateral absence needs renal US; bilateral absence needs CFTR testing and renal US.
- Palpable seminal vesicles suggest ejaculatory duct obstruction (EDO).

Varicocele grade	Definition
Subclinical	Seen only on Doppler; **do not treat**
I	Palpable only with Valsalva, standing
II	Palpable without Valsalva, not visible
III	Visible through scrotal skin

4 · Endocrine evaluation

Hormones separate treatable endocrine causes from testicular failure.

ALGORITHM

1. Take a morning sample for total T, sex hormone-binding globulin (SHBG) and albumin, and calculate bioavailable T.
2. Measure LH and follicle-stimulating hormone (FSH), then E2.
3. Measure prolactin only if indicated.

- Adequate total T is ≥ **280–300 ng/dL**. Low T is common even with density >20 million/mL, so assess every infertile man (not needed above ~66 million/mL).
- Bioavailable T is free (0.5–3%) plus albumin-bound (54–68%) T; SHBG-bound T (30–44%) is unavailable. Use the Vermeulen formula.
- Low SHBG (obesity, hypothyroidism, diabetes, steroids) makes total T under-read. High SHBG (ageing, cirrhosis, hyperthyroidism, HIV, anticonvulsants) makes it over-read.
- For borderline values, average 3 samples 20 min apart.
- The odds of abnormal concentration rise from an FSH of 4.5 IU/L. Anti-müllerian hormone (AMH) is experimental.
- A T (ng/dL) to E2 (pg/mL) ratio **<10:1** means aromatase excess.
- Prolactin of 20–50 µg/L needs no further work-up; a significant rise needs pituitary MRI.

5 · Semen analysis

The cornerstone test; its limits are references, not pass marks.

- P4 Semen analysis nomenclature — *collection, ≥2 samples 2–3 weeks apart; WHO 2021 5th centiles (volume **1.4** mL, **16** million/mL, 39 million, 42/30%, **4** % strict forms, 54% vitality); terms; only azoospermia definitive; HOS test*

Not in the paper answer:

- World Health Organization (WHO) 6th edition 50th centiles: 3.0 mL, 66 million/mL, 210 million, 64% total motility, 55% progressive, 14% normal forms, 78% vitality.
- Semen analysis discriminates poorly between fertile and infertile men (area under the curve [AUC] ≈0.6).
- The legacy WHO concentration threshold was 20 million/mL; Guzick's subfertile and fertile thresholds were 13.5 and 48 million/mL.
- Brown semen is often seen in spinal cord injury (SCI). Isolated teratozoospermia does not predict natural, intrauterine insemination (IUI) or in vitro fertilisation (IVF) outcome.
- †Sperm DNA fragmentation testing (SCSA, TUNEL) is not routine. Use it for recurrent pregnancy loss or failed assisted reproduction (ART). Varicocele, smoking and oxidative stress raise it.
- **Low volume (<1.0 mL)** suggests EDO, CBAVD, hypoandrogenism, retrograde ejaculation, neuropathy or α-blockers. Do a **postejaculatory urinalysis**, which is significant if urinary sperm roughly equal antegrade sperm. If there are no sperm in urine, do transrectal ultrasound (TRUS).

6 · Genetic testing

Genetics predicts sperm retrieval and inheritance.

ALGORITHM

1. If non-obstructive azoospermia (NOA) or density **<5 million/mL**, then do karyotype and Y-microdeletion testing (ASRM/AUA 2021).
2. If CBAVD (or idiopathic EDO), then test CFTR in the man **and his partner**.

- P3 Genetic causes of male infertility. Enumerate and classify. — *AZFa (USP9Y, DDX3Y)/AZFb complete deletion: no retrieval; AZFc (DAZ): sperm often found, sons inherit; CFTR; Kallmann; androgen receptor; globozoospermia, AURKC, PCD*

Not in the paper answer:

- AZFb genes include EIF1AY and KDM5D. Palindromic amplicons on Yq make the region fragile.
- AZFc deletion with oligozoospermia declines progressively, so **cryopreserve sperm** early.

7 · Imaging

Imaging answers specific questions only.

- Do not screen, because incidental microlithiasis causes harm. Varicocele is clinical.
- P3 Ejaculatory duct obstruction — *TRUS vesicle AP **>1.5 cm** with or without midline cyst (low specificity); aspiration ≥3 sperm/HPF; vesiculography, chromotubation; <5% of azoospermia; idiopathic EDO CFTR in ~half; TURED with TRUS guidance*

8 · Azoospermia: obstructive (OA) vs non-obstructive (NOA)

This split decides between reconstruction and sperm extraction.

	OA	NOA (spermatogenic dysfunction)
FSH	**≤7.6 IU/L**	>7.6 IU/L
Testis long axis	>4.6 cm	≤4.6 cm
Management	Reconstruction *(see Ch 70)* or sperm retrieval	Genetic tests; microdissection testicular sperm extraction (micro-TESE) with intracytoplasmic sperm injection (ICSI)

ALGORITHM

1. If the vasa are absent, then test CFTR in both partners.
2. If the vasa are present and volume is <1 mL, then do postejaculatory urinalysis. Sperm in urine means retrograde ejaculation; no sperm means TRUS for EDO.
3. If volume is normal, then use FSH and testis size to separate OA from NOA.

- NOA histology: hypospermatogenesis, maturation arrest, Sertoli cell–only.

9 · Specific causes and treatment

Some causes respond to drugs; others are bypassed by ICSI.

Klinefelter syndrome (47,XXY)

- P4 Klinefelter's syndrome. — *commonest genetic cause; 1 in 500–1000; non-mosaic 80–90%; small firm testes, raised FSH/LH, low T; ejaculated sperm ~8%; micro-TESE ~**50%**; mediastinal GCT*
- **Trap**: exogenous T suppresses residual spermatogenesis, so extract sperm before starting T. If fertility is wanted, use aromatase inhibitors or human chorionic gonadotrophin (hCG).

Hypogonadotropic hypogonadism — most treatable cause

State	Hormones	Treatment
Kallmann (anosmia; KAL1, GNRHR)/complete HH	Low LH, low FSH	hCG with or without recombinant FSH (rFSH) or human menopausal gonadotrophin (hMG)
Isolated LH deficiency ("fertile eunuch")	Low LH, normal FSH	hCG
Isolated FSH deficiency	Low FSH	FSH

State	Hormones	Treatment
Hypothalamic GnRH deficiency	Low GnRH	Pulsatile subcutaneous GnRH every 2 h, for ≥6 months
Partial HH (common)	LH inappropriately low for T	Selective oestrogen receptor modulator (SERM) or aromatase inhibitor

- Give hCG 1500–5000 IU 2–3 times weekly (max 10,000/week), titrated to T; add hMG or rFSH 75 IU 2–3 times weekly. After androgen abuse, allow 1–2 years.
- Treat prolactinoma with cabergoline. Treat CAH with corticosteroids and watch for testicular adrenal rest tumours.

Empirical medical therapy

Drug	Dose	Notes
Clomiphene (SERM)	25 mg daily or 50 mg alternate days; max 100 mg	Blocks E2 feedback, so LH and FSH rise; off-label; no change in prostate-specific antigen (PSA) or haematocrit; target T 600–800 ng/dL
Anastrozole / letrozole	Anastrozole 1 mg daily	Raises T and lowers E2; use when T:E2 <10:1 (obesity); caution with bone density

Primary failure and androgen resistance

- High LH with low T has no accepted therapy; exogenous T is contraindicated, so offer sperm extraction.
- Androgen receptor defects (Xq11–12, long CAG repeat) give high T, LH and E2; treat with ICSI. ICSI pregnancies are reported in 5α-reductase type 2 deficiency.

CBAVD

- P3 CBAVD — *genital form of CF; mild CFTR alleles; normal spermatogenesis, low-volume azoospermia; partner carrier ~1 in 20; renal agenesis ~1 in 10; retrieval + ICSI*

Varicocele

- P3 Varicocele: controversies and present status of treatment — *prevalence, left predominance, right-sided: exclude tumour; heat/ROS/DNA injury; repair palpable + abnormal semen, never subclinical; may down-stage ART*
- Varicocele affects ~15–20% of all men versus 35–50% of infertile men *(see Ch 70)*.

Antisperm antibodies and structural defects

- Antisperm antibodies: washing fails and steroid trials conflict, so use **ICSI**.
- Globozoospermia: ICSI with or without oocyte activation. Macrocephalic multiflagellate sperm (AURKC): ICSI not recommended. Primary ciliary dyskinesia (PCD): ICSI, testicular sperm better.

10 · Lifestyle and obesity

Lifestyle change helps every treatment.

- In obesity, more aromatase raises E2, which lowers LH and T; SHBG and inhibin B fall. Hormone-independent DNA effects mean correcting hormones may not restore fertility. Obesity worsens ART outcomes, so advise weight loss.
- Stop smoking, cannabis, excess alcohol and anabolic steroids; avoid heat; offer psychological support.

11 · Assisted reproduction

Sperm count decides the technique.

- Preparation is by density gradient or swim-up. ICSI needs one sperm per oocyte and bypasses selection.
- †Choose by total motile sperm count (TMSC): IUI if processed TMSC is ≥5–10 million; IVF below that or after failed IUI; ICSI for severe oligozoospermia, retrieved sperm, antisperm antibodies or prior fertilisation failure.
- †Retrieval in OA is by percutaneous or microsurgical epididymal sperm aspiration (PESA/MESA) or testicular aspiration; in NOA it is micro-TESE.

70 Surgical Management of Male Infertility

1 · Principles

Why this matters: treating the man is cheaper and safer than assisted reproduction.

- In vitro fertilisation (IVF) with intracytoplasmic sperm injection (ICSI) treats even non-obstructive azoospermia (NOA). But it is costly, burdens the female partner and bypasses natural barriers (genetic transmission, birth defects).
- Treating the man (reconstruction for obstructive azoospermia (OA), varicocelectomy) is safest and cheapest, and moves the couple **down the ladder**: donor sperm, then IVF/ICSI with testicular sperm, then IVF/ICSI with ejaculated sperm or intrauterine insemination (IUI), then natural conception.

2 · Surgical anatomy

Why this matters: reconstruction fails without blood supply.

Structure	Arteries (origin)
Testis	**Testicular** (aorta) — main; deferential (internal iliac/superior vesical); cremasteric (inferior epigastric; mainly tunica vaginalis)
Epididymis	Superior/medial epididymal (from testicular a.) supply the caput; inferior epididymal (from deferential a.) supplies the cauda; interconnected
Vas	Abdominal end: deferential only. Testicular end: deferential and inferior epididymal. **No supply from cremasteric or cord vessels**

- The testicular artery **exceeds deferential and cremasteric combined**. After vasectomy the vasal artery is presumed lost, so preserve the testicular artery at any later surgery.
- The epididymis can be dissected off the testis (inferior/medial epididymal arteries ligated) if the **superior epididymal artery is intact**; this gains **4–6 cm**.
- If the vas is blocked at two sites with vessels cut at both, the segment between fibroses. So **never do two simultaneous vasovasostomies on one vas**; excise it and do one tension-free anastomosis.
- **Excurrent ducts**: the rete drains via 7–11 efferent ducts into the caput, and a **single tubule** runs from the caput–corpus junction to the vas. So injury distal to the caput (even **one stitch**) obstructs the whole side; a single puncture is safe only at the most proximal caput.

3 · Diagnostic testis biopsy

Why this matters: biopsy separates obstruction from failed production and banks sperm.

- P1 Testicular biopsy and TESA (testicular sperm aspiration) — *indications (azoospermia, normal testes, palpable vasa, normal FSH); diagnostic + therapeutic; open = gold standard; never formalin; percutaneous 14 G / 23 G*

Not in the paper answer:

- Biopsy is usually bilateral and planned as testicular sperm extraction (TESE), with an andrology laboratory on site for wet examination and freezing.
- Avoid the blind 14-G core in a scarred scrotum.

ALGORITHM

1. Give a cord block; choose the larger testis (opposite a varicocele).
2. Make a 2 cm incision, open the tunica vaginalis, place stay sutures.
3. Incise the albuginea 3–4 mm, using bipolar on crossing vessels.
4. Take a pea-sized sample with iris scissors (no forceps).
5. Do a touch prep, then a wet prep. If negative, then biopsy the other side.
6. Close the tunica vaginalis with continuous 5-0; no antibiotics.

- With the microscope at 10–25×, sample the larger tubules. A non-absorbable suture marks a positive site for future TESE.

Complication	Point
Haematoma — most common	Prevent with magnification, bipolar and continuous tunica vaginalis closure
Epididymal biopsy — most serious	If that tubule held sperm, obstruction is now certain

4 · Vasography

Why this matters: it locates a block, but careless technique causes stricture.

- **Absolute indications (all three)**: azoospermia, complete spermatogenesis on biopsy and at least one palpable vas. Relative: severe oligospermia with normal biopsy, high sperm-bound antibodies, and low volume with poor motility (partial ejaculatory duct obstruction (EDO)).
- Do it **only at planned reconstruction**, because of scar and stricture. It cannot diagnose epididymal obstruction, the commonest non-vasectomy site.

ALGORITHM

1. High scrotal incision; find the vas at the straight–convoluted junction; hemivasotomy at 25×.
2. **Examine vasal fluid first.** If motile sperm, then barbotage and cryopreserve **before any dye**.
3. Pass a 24-G sheath into the abdominal end and inject 1 mL saline. If flow is easy, then it is patent.
4. Rarely, give dilute indigo carmine; blue urine confirms.
5. Use formal contrast only for obstruction above the internal ring.
6. Close with 10-0 mucosa and 8-0 muscularis.

Vasal fluid / vasogram	Meaning and action
No sperm, abdominal end patent	Epididymal obstruction: vasoepididymostomy
Copious fluid, many sperm	Vasal obstruction or EDO: locate it (vasogram, or 2-0 Prolene threaded down)
Block at ejaculatory ducts; both vasa fill from one side	EDO or midline cyst: transurethral resection of the ejaculatory ducts (TURED)
Vas ends blindly	Partial vasal agenesis: cystic fibrosis gene (CFTR) testing; retrieve and freeze
Inguinal block	Inguinal or crossed vasovasostomy
Many sperm, normal vasogram	Retrograde ejaculation, emission failure, aperistalsis

- Use **indigo carmine, not methylene blue**, which kills sperm (use it only after banking).
- Complications are **stricture** (repeated punctures, poor closure, non-water-soluble contrast), devascularisation, haematoma and granuloma.
- **Transrectal seminal vesiculography**: dilated seminal vesicles are aspirated under transrectal ultrasound (TRUS). Sperm mean at least one vas and epididymis are patent, so contrast and indigo carmine allow TURED without scrotal exploration. No sperm means secondary epididymal obstruction, so plan retrieval and ICSI.

5 · Vasovasostomy (vasectomy reversal)

Why this matters: vasal fluid decides the operation on the table.

Evaluation

- Prior fertility proves spermatogenesis. Check follicle-stimulating hormone (FSH) if testes are small or soft; raised FSH means poorer prognosis.
- An indurated epididymis predicts vasoepididymostomy. A **sperm granuloma at the testicular end gives a better prognosis**, because it vents pressure, whatever the interval. A large gap needs inguinal extension; groin scars suggest iatrogenic obstruction.
- Whole sperm in the centrifuged pellet mean sperm are certain in at least one vas. Low volume needs TRUS.
- Use general anaesthesia; operate only if able to do vasoepididymostomy.

Approach and preparation

- Approach: scrotal for most; inguinal for hernia or orchidopexy obstruction; robotic above the inguinal ligament.
- **Never strip the perivasal sheath**; excise a granuloma staying on the vas.
- **Length**, in escalating steps: dissect to the external then internal ring; free the convoluted vas (4–6 cm; **do not unravel**); reroute beneath the inguinal floor; take the epididymis off the testis. This bridges gaps up to **10 cm**.
- Cut through a slotted clamp. A healthy end has a white mucosal ring, soft muscularis and a bleeding edge. **If the muscularis is gritty or bleeding poor, recut.**
- At the testicular end **aspirate, never flush**. Flush saline into the abdominal end to prove patency.

Intraoperative fluid grading — the decision

Vasal fluid	Microscopy	Operation
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin	Sperm with tails	Vasovasostomy
Copious, creamy yellow	Many sperm heads	Vasovasostomy
Scant, granuloma present	Barbotage: sperm	Vasovasostomy
Thick white **toothpaste**	No sperm/heads	**Vasoepididymostomy**
Scant white thin, or dry without granuloma	No sperm/heads	**Vasoepididymostomy**
Any	Grape-like clusters of heads	**Vasoepididymostomy**

KEY POINT

Clear watery fluid still gets vasovasostomy: the epididymis has not blown out, and sperm usually return. Toothpaste fluid means secondary epididymal blow-out, so open the tunica vaginalis; a discrete level (dilated above, collapsed below) confirms it. When in doubt or inexperienced, do vasovasostomy.

Anastomotic principles

Principle	Reason
Accurate mucosa-to-mucosa	Testicular lumen is 2–5× the abdominal one
Leakproof	Sperm are antigenic; fluid lacks clotting factors, so the seal depends on mucosal sutures
Tension-free	Tension gives early sperm, then falling counts, then azoospermia
Good blood supply; healthy mucosa/muscularis; atraumatic	Recut until bleeding and soft; redo if back-walled or torn

- **Multilayer microdot**: six mapped dots guide **six 10-0 mucosal sutures** (double-armed, inside-out), then 9-0 deep muscularis, 9-0 adventitia and 8-0 sheath. Two-layer alternative: 10-0 mucosa and 8-0 full thickness.
- **Reinforcing vasal (ReVas)**: the sheath 3 cm either side is looped with 5-0 Prolene plus 5–10 reinforcing sutures, relieving tension and raising patency.
- **Convoluted vas**: cut transversely, do not unravel, take small bites and reinforce.
- **Crossed vasovasostomy** suits unilateral inguinal obstruction with a contralateral atrophic testis, or one unreconstructable side. Transpose the testis if the vas is very short.
- **Varicocele with reversal**: vasal vessels are compromised, so stage varicocelectomy ≥6 months later.

Post-operative care and outcomes

- Scrotal support and no heavy lifting for 4 weeks; ejaculation at 2 weeks, intercourse at 4 weeks.
- Check semen at **6 weeks**, then 6 months if no pregnancy. Poor motility gets a 4-week prednisone course. Azoospermia at 6 months means redo.
- Perivasal haematoma is commonest; an anastomotic granuloma heralds obstruction. **Late stricture** shows as falling motility, then count. **Bank sperm once motile sperm appear.**
- With sperm in vasal fluid on at least one side, sperm return in virtually all. Pregnancy is about half, depending on vasectomy interval and female age.

6 · Vasoepididymostomy

Why this matters: it is needed when the block is epididymal.

- The epididymis is a single tubule ~200 µm wide and 12–15 feet long, with its thinnest wall at the caput. So a **more distal anastomosis is easier and gives better pregnancy**. Adaptation continues up to 2 years.
- **Indications**: reversal by the fluid criteria; or non-vasectomy OA with proven spermatogenesis, no vasal sperm and a patent vas and ejaculatory ducts.

ALGORITHM

1. Hemitransect the vas at the straight–convoluted junction; sample fluid; confirm patency.
2. Open the tunica vaginalis; choose a site **above** the obstruction where tubules are dilated. If unclear, then puncture distal to proximal until sperm appear.
3. Make a 3–4 mm tunic window.
4. Bring the vas through the tunica vaginalis, **fixed with 5-0 Prolene to the albuginea (tension relief)**.
5. Place posterior tunic–vas 8-0, then intussusception sutures, then outer tunic–vas adventitia.

Technique	For	Against
Longitudinal intussusception (LIVE) — used for all	Two 10-0 longitudinal sutures placed in the **still-distended** tubule; incise between; tubule pulled into vas; bloodless, leakproof	Fluid not checked before sutures set
End-to-side	Bloodless; fluid checked first	Suturing a collapsed tubule
End-to-end	Rapid level finding; bridges gap	Haemostasis hard; inferior epididymal supply lost

- Intussusception works because needles enter before collapse, and forward flow pushes the tubule against vasal mucosa.
- **No simultaneous varicocelectomy**; stage it after 6 months.
- Late shutdown occurs, so **bank sperm during surgery and once they appear**. If it fails, redo or retrieve sperm for ICSI.

7 · TURED

Why this matters: EDO is a correctable cause of low-volume azoospermia.

- P3 Ejacultory duct obstruction — *congenital (midline cyst, atresia) vs acquired; low volume, acidic, low fructose, palpable vas, normal FSH; TRUS; TURED only after couple declines IVF; sperm return in ~half; cold-knife reobstructs; reflux, epididymitis, retrograde ejaculation; vesiculoscopy*

Not in the paper answer:

- Use minimal coagulation and give 7 days of antibiotics. Avoid confirmatory high-pressure vesiculography, which can cause secondary epididymal obstruction.
- Document reflux by voiding cystourethrogram (VCUG) or semen creatinine.
- Treat epididymitis with low-dose antibiotics until pregnancy; if chronic, then vasectomy or epididymectomy.
- For retrograde ejaculation give pseudoephedrine 60 mg 90 min before, or use alkalinised urinary sperm.

ALGORITHM

1. Use a 24-Fr resectoscope with a finger in the rectum.
2. Resect the **verumontanum** to open the dilated orifices or cyst, sparing bladder neck, sphincter and rectum.
3. **Blue efflux confirms** adequacy; coagulate minimally.
4. Catheter overnight; 7 days of antibiotics.

8 · Sperm retrieval

Why this matters: when reconstruction is impossible, retrieval with ICSI is the only route.

- P2 Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction) — *ICSI essential; PESA/MESA/TESA/TESE/micro-TESE table; MESA by capillary action; micro-TESE ~half of NOA; retrieval by diagnosis (AZFa/b **0% — do not operate**); frozen = fresh except SCI*

Not in the paper answer:

- Percutaneous epididymal sperm aspiration (PESA) gives pregnancy about half that of open methods, often with too few sperm to freeze.
- Open TESE is the most consistent open technique. TESA may need 10–20 passes.
- In spinal cord injury (SCI), fresh testicular sperm may beat frozen.
- Microsurgical epididymal sperm aspiration (MESA) gives most motile sperm and multiple vials. **Microdissection TESE (micro-TESE) is best in NOA**.

KEY POINT

Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved.

ALGORITHM

1. Micro-TESE: general or regional anaesthesia.
2. At 10×, find the avascular plane on the anterior albuginea; make a wide equatorial incision and bivalve the testis.
3. Excise larger tubules; an andrologist in theatre examines them; map positives.
4. If negative, then explore the other testis.
5. Close the albuginea with 5-0 and the tunica vaginalis with 4-0.

9 · Cryopreservation

Why this matters: every operation is a chance to bank sperm.

- Freeze at diagnostic biopsy, at every reconstruction, and from the ejaculate once motile sperm return after reversal. Sample before any dye; epididymal vials allow elective retrieval.

10 · Varicocelectomy

Why this matters: technique decides recurrence and hydrocele.

- P2 P3 Surgical management of bilateral varicocele; Varicocele: controversies — *indications; approach table (microsurgical recurrence <1%, hydrocele ~0); micro-Doppler, testis delivery, impulse test; complications*

Not in the paper answer:

- Conventional inguinal repair: hydrocele 3–15%, recurrence 5–15%, because it misses scrotal and periarterial veins.
- Subinguinal repair means less pain but more veins and a branched artery. **Open the external oblique** in prepubertal children, solitary testis, tight low ring, short cord, inexperience or hernia repair; stay subinguinal after inguinal surgery or in obesity.
- Pulsation is unreliable; the artery (1–1.5 mm, ~two-thirds of flow) often lies under a large vein.
- Ligate vasal veins only if >2.5 mm, keeping one set: **one intact deferential set guarantees venous return**.

- Aftercare: intercourse at 1 week; activity at 4 weeks (6 if fascia opened). Treat recurrence by venography and embolisation.
- Repair halts decline and improves semen, live birth, IVF/ICSI outcomes and testosterone; younger men gain more.

ALGORITHM

1. Local anaesthesia with sedation; 2–3 cm incision over the external ring.
2. Deliver the cord on a Penrose, excluding the ilioinguinal nerve and genital branch.
3. **Map arteries with micro-Doppler before handling.**
4. Open the spermatic fasciae; work at 10–25× with 1% papaverine; loop the artery.
5. Ligate **all veins except vasal**; preserve lymphatics.
6. Deliver the testis; ligate external spermatic and gubernacular veins.
7. Do the impulse test. If there is no impulse, then a vein was missed.

- **End point**: only testicular and cremasteric arteries, lymphatics, and the vas with its vessels remain.

KEY POINT

Recurrence means a vein was missed; hydrocele, a lymphatic tied; atrophy, an artery tied. Microsurgery with testicular delivery addresses all three.

11 · Landmark trials

Why this matters: this trial justifies repairing clinical varicocele.

Trial	Design	Result	Take-home
Madgar (1995)	Randomised controlled trial (RCT): varicocelectomy vs no surgery in infertile men	1-year pregnancy 44% vs 10%	Repair of clinical varicocele with abnormal semen improves fertility

71 Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

1 · Definition and key concepts

Why this matters: every ED treatment targets one step of this mechanism.

- **Erectile dysfunction (ED)** is the consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction (American Urological Association (AUA) 2018). The term replaces "impotence".
- Erection needs **sinusoidal (trabecular) smooth muscle relaxation, arterial dilation and venous compression (veno-occlusion)**. It is a neurovascular event: rigidity comes from *trapping* blood, not from inflow alone.
- Milestones MCQ: Leonardo da Vinci (hanged men) showed erection is blood; the 1990s identified nitric oxide (NO); 1998 brought oral phosphodiesterase type 5 inhibitors (PDE5Is).

2 · Functional anatomy

Why this matters: the tunica and vessels explain veno-occlusion, penile fracture and implant perforation.

Corpora

- There are paired dorsal **corpora cavernosa** and a ventral **corpus spongiosum**, which surrounds the urethra and expands as the glans.
- The crura arise separately from the puboischial rami. The septum is **incomplete** in humans, so one injection fills both corpora.
- Sinusoids (larger centrally) are separated by smooth muscle trabeculae. Helicine arteries and terminal cavernous nerves lie close to this muscle.
- Sinusoidal oxygen tension (Po_2) is venous when flaccid (**~35 mm Hg**) and arterial when erect (**~90 mm Hg**).

Tunica albuginea

- The **inner circular layer** holds the cavernous tissue and gives off **intracavernous pillars** (absent at 5–7 o'clock). The **outer longitudinal layer** runs from glans to crura; it is thicker dorsally and absent in the ventral groove.
- The ventral tunica is ~0.8 mm against ~2.2 mm dorsally. So 5–7 o'clock is the weak zone for **penile fracture** and implant crossover or perforation.
- **Veno-occlusion**: emissary veins run obliquely between the layers and are compressed as the sinusoids stretch the tunica. Cavernous arteries escape because they lie in a periarterial sheath.
- The spongiosal tunica is thin and single-layered, has no struts and is absent in the glans. So spongiosum and glans are **low-pressure arteriovenous shunts** and never become rigid.

Arteries

- The internal iliac gives the **internal pudendal**, which becomes the common penile artery with three branches.
- The **cavernous** artery is the main corporal supply. Its **helicine arteries** are tortuous when flaccid and straight when erect, and open into sinusoids without capillaries.

- The **dorsal** artery fills the glans (its perforators join the cavernous artery distally). The **bulbourethral** artery supplies the bulb and spongiosum.
- **Accessory pudendal arteries** (from external iliac, obturator, vesical or femoral) occur in ~one-third and may be dominant. Preserving them at radical prostatectomy speeds erectile recovery.

Veins

Sinusoids drain into the **subtunical venous plexus**, which exits through **emissary veins** in the tunica.

System	Drains	Destination
Deep dorsal vein (with or without circumflex and periurethral veins)	Glans and distal two-thirds of corpora	Periprostatic (Santorini) plexus
Cavernous and crural veins	Proximal (infrapubic) penis	Internal pudendal veins
Superficial dorsal vein	Skin, subcutaneous tissue	Saphenous veins

> **KEY POINT**
> Arterial and venous anatomy vary widely between men, so the same "nerve-sparing" or pelvic operation gives different erectile outcomes.

3 · Haemodynamics: phases of erection and detumescence

Why this matters: phase timing explains penile Doppler readings and how detumescence fails in priapism.

- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *sequence of events and phase table (flaccid, latent, tumescence, full, rigid, detumescence)*

Not in the paper answer:

- Full-erection intracavernous pressure (ICP) plateaus at **~100 mm Hg**; the rigid phase is suprasystolic. The cavernous (autonomic) nerve drives phases 1–3, and the pudendal nerve, via ischiocavernosus, drives phase 4.
- In the rigid phase arterial inflow is near zero or reversed.
- Detumescence has three phases: 5, initial (a transient ICP rise as muscle contracts against closed veins); 6, slow (veins gradually reopen); 7, fast (outflow restored).
- **Trap**: arterial inflow peaks in the latent phase, not at rigidity. This is why peak systolic velocity (PSV) on Doppler is measured early.
- Spongiosum and glans lack veno-occlusion, so their pressure is one-third to one-half of corporal. In the rigid phase they engorge further from bulbo- and ischiocavernosus contraction and dorsal vein compression under Buck fascia.

4 · Neuroanatomy

Why this matters: knowing the nerve pathways predicts erections after spinal injury and pelvic surgery.

Peripheral and spinal

	Sympathetic	Parasympathetic	Somatic
Origin	**T11–L2** (often T10–T12)	**S2–S4** intermediolateral columns	S2–S4; motor from **Onuf nucleus**

	Sympathetic	Parasympathetic	Somatic
Route	Sympathetic chain, superior hypogastric plexus, hypogastric nerves, then pelvic plexus	Pelvic nerves (nervi erigentes) to the pelvic plexus	Pudendal nerve; afferent via dorsal nerve of penis
Role	**Detumescence**; emission	**Erection** (cavernous nerves)	Sensation; ischiocavernosus (rigidity), bulbocavernosus (ejaculation)

- **Cavernous nerves** arise from the pelvic plexus and carry both autonomic types. They †run posterolateral to the prostate; the lateral branch pierces the urogenital diaphragm 4–7 mm lateral to the sphincter.
- Pelvic ganglia lie along the bladder–prostate junction and seminal vesicles, so they are vulnerable in radical pelvic surgery.
- The dorsal nerve is somatic sensory but carries some neuronal NO synthase (nNOS) fibres.
- **Bulbocavernosus reflex**: squeezing the glans makes the bulbocavernosus contract. It has limited value in ED work-up.

Three types of erection

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Psychogenic	Audiovisual, fantasy	Brain modulates spinal centres, chiefly by suppressing sympathetic tone	May persist with sacral lesions; absent with lesions above T9
Reflexogenic	Genital touch	Sacral reflex arc to the cavernous nerves	Preserved in upper cord lesions
Nocturnal	Rapid eye movement (REM) sleep	Cholinergic pontine tegmentum on; locus coeruleus (noradrenergic) and raphe (serotonergic) silent	—

Supraspinal (central)

- The **medial preoptic area (MPOA)** handles partner recognition and integrates hormonal and sensory cues; it is dopaminergic.
- The **paraventricular nucleus (PVN)** sends pro-erectile **oxytocin** neurons to the lumbosacral nuclei. The nucleus paragigantocellularis is the **serotonergic brake**.
- Pyriform cortex lesions cause hypersexuality.

Central neurotransmitters

- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *promoters (dopamine, oxytocin, NO, noradrenaline, melanocortins, 5-HT2C) vs inhibitors (5-HT1A/2, GABA, opioids, cannabinoids, prolactin)*

Not in the paper answer:

- Apomorphine is a D1/D2 agonist; it acts in the PVN, which releases oxytocin and then NO. Melanocortins act on the MC4 receptor.
- Serotonin 5-HT1A and 5-HT2 receptors inhibit erection but facilitate ejaculation.
- Arousal deactivates temporal and parietal cortex. In psychogenic ED the cingulate and frontal cortex stay active, and apomorphine normalises this.

KEY POINT

NO synthase (NOS) inhibitors block both apomorphine- and oxytocin-induced erections, so NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally.

5 · Molecular mechanisms of smooth muscle tone

Why this matters: every erectogenic drug acts on one of these pathways.

Contraction (flaccid state)

- Corporal smooth muscle is **contracted at rest** through myogenic tone, adrenergic tone and endothelial constrictors.
- **Calcium pathway**: noradrenaline (α1), endothelin-1 (ET_A), angiotensin II (AT1), PGF2α and thromboxane A2 act via Gq and phospholipase C to form inositol trisphosphate (IP_3). This releases and admits Ca^{2+}; Ca^{2+}–calmodulin then activates **myosin light chain (MLC) kinase**, which phosphorylates MLC20.
- **Ca^{2+} sensitisation by RhoA/Rho-kinase**: it inhibits **MLC phosphatase (MLCP)**, giving sustained contraction *without* a further Ca^{2+} rise. The latch state (calponin) keeps basal tone cheaply.
- Rho-kinase is raised in diabetes mellitus (DM), hypertension, hypercholesterolaemia, hypogonadism, ageing and nerve injury. It also drives fibrosis by turning fibroblasts into myofibroblasts.

Relaxation (erection)

- **NO–cGMP**: NOS converts L-arginine and O_2 into NO and citrulline. **Neuronal NOS initiates and endothelial NOS (eNOS) sustains** erection.
- NO activates soluble guanylyl cyclase (haem-dependent) to make cyclic GMP (cGMP), which activates **protein kinase G (PKG-I)**. PKG causes K^+ efflux, lowers Ca^{2+} entry and release, inhibits Rho-kinase and activates MLCP, so the muscle relaxes.
- **cAMP pathway**: PGE1 (via EP2/EP4), vasoactive intestinal peptide (VIP) and calcitonin gene-related peptide (CGRP) act via Gs to raise cyclic AMP (cAMP), which activates protein kinase A (PKA). It is **NO-independent**, which is why intracavernosal alprostadil works when nerves or endothelium fail. Prostacyclin (PGI2) contracts; PGE1 relaxes.
- **PDE5 (cGMP-specific) is the principal cavernous PDE**. PDE6 is retinal (visual side effects); PDE2, 3 and 4 degrade cAMP. Papaverine is nonspecific. Raised cGMP inhibits PDE3, which raises cAMP.
- K^+ channels (maxi-K, K_{ATP}) hyperpolarise the cell. **Gap junctions (connexin-43)** synchronise relaxation, so a local input gives a uniform erection.
- Oxygen acts as a switch: low flaccid Po_2 suppresses NOS and favours endothelin, while erect Po_2 >50–60 mm Hg permits full NOS activity.
- **Adenosine** (A2 receptor, raising cAMP): excess adenosine with increased A2B signalling causes prolonged erection in adenosine deaminase-deficient and sickle cell mice, the **priapism models**.
- **Sonic hedgehog (SHH)** maintains corporal architecture. It falls after nerve injury and in DM, causing atrophy.

- Acetylcholine acts indirectly (endothelial NO, presynaptic adrenergic inhibition), so **atropine does not abolish** neural erection MCQ.
- PKG-I knockout mice cannot erect, but nNOS or eNOS knockouts often can (redundancy). So PKG is the bottleneck.

Second messenger	Activators	Kinase	Degraded by	Drug
cGMP	NO, CO, H_2S, natriuretic peptides	PKG-I	PDE5	PDE5Is
cAMP	PGE1, VIP, CGRP, adenosine (A2)	PKA	PDE2, 3, 4	Alprostadil; papaverine (both)

Detumescence

NO release stops and PDE5 hydrolyses cGMP. Sympathetic noradrenaline (α1; the principal transmitter of flaccidity), endothelin-1 and Rho-kinase then contract the trabeculae, so the veins reopen (phases 5–7).

> **KEY POINT**
> Corporal tone is a tug-of-war over one phosphorylation site (MLC20): Ca^{2+}–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" means the phosphatase.

6 · Classification of ED

Why this matters: classifying by mechanism tells you which test and treatment fit.

Approach	Categories
Neurovascular mechanism	Failure to **initiate** (neurogenic, psychogenic, endocrine) · to **fill** (arterial) · to **store** (veno-occlusive)
Aetiological	Psychogenic · neurogenic · hormonal · arteriogenic · cavernosal (venogenic) · drug-induced · systemic disease · mixed (commonest)

Most organic ED has a psychogenic component.

7 · Risk factors and endothelial dysfunction

Why this matters: ED often signals vascular disease before the heart does.

- **Age** is the strongest demographic predictor, but DM, heart disease and hypertension carry more risk (Massachusetts Male Aging Study (MMAS)).
- **Endothelial dysfunction**: oxidative stress oxidises tetrahydrobiopterin (BH4), causing **NOS uncoupling**, so superoxide forms peroxynitrite. Arginase depletes L-arginine and asymmetric dimethylarginine (ADMA) inhibits NOS. Together these lower NO.
- **Artery-size hypothesis**: smaller penile arteries become symptomatic first. ED preceded coronary artery disease (CAD) by a mean of **~2 years**, so ED is a **sentinel** and needs cardiac risk stratification *see Ch 72*.
- The common end-path is raised Rho-kinase and transforming growth factor-β (TGF-β). These add collagen and lower the smooth muscle-to-collagen ratio, causing veno-occlusive failure.

8 · Pathophysiology by type

Why this matters: each cause damages a different step, which guides work-up and treatment.

Psychogenic

- The brain inhibits the spinal centres, and excess sympathetic tone (higher serum noradrenaline) contracts the muscle.
- Causes include performance anxiety, partner conflict, depression, post-traumatic stress disorder (PTSD) and problematic pornography use.

Neurogenic

- **Brain**: Parkinson disease, stroke, tumours, dementia, temporal lobe epilepsy, multiple system atrophy and head injury. **Cord**: SCI (commonest), spina bifida, disc disease and multiple sclerosis (MS).

Complete lesion	Reflexogenic erection	Psychogenic erection
Upper cord (above sacral centre)	Preserved in **~95%**; brief, need continuous stimulation	Lost
Lower cord (sacral)	Preserved in only **~25%**	May occur (thoracolumbar pathway); marginal tumescence

- **Iatrogenic**: radical prostatectomy, cystoprostatectomy, abdominoperineal resection (APR) and pelvic radiation. Most men have at least a transient decline despite nerve sparing, worse with baseline ED and age >60. Retroperitoneal sympathetic injury causes loss of emission.
- Cavernous nerve injury lowers nNOS and SHH and causes apoptosis and fibrosis, leading to venous leak. This is the rationale for early penile rehabilitation.
- **Pelvic fracture** injures the cavernous nerves or pudendal artery. With posterior urethral injury, early realignment preserves potency better than delayed anastomosis.

Hormonal

- **Testosterone (T)** governs **libido** more than mechanics, but it maintains NOS, PDE5 and smooth muscle. Castration reduces inflow and causes venous leak. Low libido is the commonest sexual symptom of low T.
- AUA: measure **morning, fasting** T in all men with ED. Low T is **<300 ng/dL**; treat only with ≥2 low levels and symptoms. Nocturnal erections need ~200 ng/dL.
- Nocturnal erections are androgen-dependent, but erections to visual stimuli are largely independent MCQ. Dihydrotestosterone (DHT) is the key penile androgen.
- **Hyperprolactinaemia** (prolactinoma, antipsychotics) suppresses gonadotrophin-releasing hormone (GnRH), causing low T, low libido and less MPOA dopamine. Low prolactin (<3 ng/mL) is also linked to ED.
- A low T:oestradiol ratio (obesity), hyperthyroidism and hypothyroidism all cause ED.

Vasculogenic – arteriogenic

- Atherosclerotic or traumatic occlusion of the hypogastric–pudendal–cavernous tree lowers perfusion pressure, giving a slower, softer erection.
- Risks are hypertension, hyperlipidaemia, smoking, DM, obesity, inactivity, pelvic irradiation and blunt perineal trauma. Trauma causes focal stenosis in young men, who are revascularisation candidates.
- Chronic hypoxia raises TGF-β1 and collagen and reduces smooth muscle. So "inflow failure becomes storage failure".
- In hypertension, angiotensin receptor blockers (ARBs) and nebivolol protect penile tissue; hydralazine and amlodipine lower blood pressure without protection.

Vasculogenic – veno-occlusive (venous leak)

- **Tunical** causes: ageing, DM, Peyronie disease, penile fracture and Peyronie surgery.
- **Fibroelastic trabecular change** (ageing, DM, nerve injury, hypercholesterolaemia) means the sinusoids cannot expand enough.
- **Insufficient relaxation** comes from anxiety, adrenergic tone or endothelial dysfunction.
- **Acquired shunts** follow priapism shunt surgery (glans–cavernosum, cavernosum–spongiosum). Congenital ectopic or crural veins cause primary ED.

> **KEY POINT**
> Veno-occlusion is an emergent property of healthy tissue (elastic tunica and fully relaxed trabeculae). So most "venous leak" is really cavernosal smooth muscle disease, which is why venous ligation surgery fails and is investigational.

- Vasculogenic ED changes: higher cavernous artery resistance; smooth muscle with higher tone, lower content and altered K^+ channels and gap junctions; cavernous fibrosis and elastin loss; impaired endothelium-dependent relaxation; altered tunical fibres; and less nNOS and eNOS.

Metabolic and systemic disease

- **DM** causes the earliest, most severe ED through arteriopathy, nitrergic neuropathy, endothelial dysfunction, advanced glycation end products (AGEs), raised Rho-kinase and arginase, and hypogonadism. **Key**: the NO *source* is damaged but the response to exogenous NO is preserved, so PDE5Is work until fibrosis sets in.
- **Metabolic syndrome (MetS) and obesity**: visceral fat is endocrine. Weight loss, Mediterranean diet, exercise and bariatric surgery help.
- **Hyperlipidaemia**: oxidised low-density lipoprotein (LDL) inhibits NO. Statins are linked to better International Index of Erectile Function (IIEF) scores.
- **Chronic kidney disease**: up to **80%** of men with end-stage renal disease have ED, and it persists in ~half after transplant.
- Others: HIV, cirrhosis, chronic obstructive pulmonary disease, obstructive sleep apnoea, and benign prostatic hyperplasia with lower urinary tract symptoms.
- **Ageing** brings longer latency, less rigidity, weaker ejaculation, a longer refractory period and fewer nocturnal erections.

Drug-induced

Class	Cause ED	Mechanism / note	Alternative
Antihypertensives	**Thiazides**, **nonselective β-blockers**, clonidine	Thiazide – unknown; β-blocker – prejunctional α2	ACE inhibitors, ARBs, nebivolol, α1-blockers (ejaculatory dysfunction), calcium channel blockers
Antidepressants	Selective serotonin reuptake inhibitors (SSRIs; paroxetine worst), tricyclics	5-HT_2/5-HT_3 spinal inhibition, less dopamine and NO; post-SSRI dysfunction may persist	Bupropion, mirtazapine, nefazodone, buspirone; add a PDE5I
Antipsychotics	Typical > atypical	Dopamine block causes hyperprolactinaemia	Newer agents

Class	Cause ED	Mechanism / note	Alternative
Antiandrogens	LHRH agonists/antagonists, androgen receptor antagonists, cyproterone, **5α-reductase inhibitors (5ARIs)**	5ARI: least T effect yet persistent ED after stopping	None
Recreational	Tobacco (dose–response, ≥20 pack-years), heavy alcohol, opioids	Endothelial damage, neuropathy, hypogonadism	Stop or moderate
Others	Cimetidine, spironolactone, carvedilol	—	—

Trazodone (via m-chlorophenylpiperazine) causes priapism. Among anticonvulsants, valproate lowers desire, and carbamazepine and phenytoin are worse than lamotrigine.

Primary (lifelong) ED

- It is usually psychogenic, but exclude hypoplastic cavernous arteries, congenital venous leak, corporal atresia and megalourethra.
- Micropenis (stretched length >2.5 SD below the mean) and curvature may present as "ED".

> **KEY POINT**
> Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction and TGF-β-driven fibrosis.

72 Evaluation and Management of Erectile Dysfunction

1 · Definition and significance

Why this matters: ED is common and often the first sign of vascular disease.

- **Erectile dysfunction (ED)** (American Urological Association, AUA): a consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction. Minimum **3 months**, except after trauma or surgery.
- About **20%** of adult men have ED (up to 15% at 40–49 y; up to 40% at 60–69 y). One in four new presenters is under 40; half of these are severe.
- Opioid use doubles the risk; prior COVID-19 raises it about sixfold.
- **ED predicts cardiovascular disease (CVD)** through shared endothelial dysfunction. It predicts as strongly as smoking or a family history of myocardial infarction (MI), and is linked to all-cause mortality. Younger onset means greater cardiovascular (CV) risk.

Independent association (multivariate)	Risk factors
Strongest	Antidepressant use, then antihypertensive use
Strong	Diabetes mellitus (DM), the strongest disease; obstructive urinary symptoms
Moderate	Hypertension, prostate enlargement, smoking, body mass index (BMI) ≥30, inactivity
Weak/null after adjustment	CVD, hypercholesterolaemia

- **Drug-induced ED** makes up to 25% of presentations *(see Ch 71)*.

2 · Management principles

Why this matters: the patient chooses treatment once his cardiac risk is known.

- **Shared decision making** (AUA 2018): all non-contraindicated options are offered **in parallel, not ranked**, and the patient may choose any first. They are phosphodiesterase type 5 inhibitors (PDE5Is), the vacuum erection device (VED), intraurethral (IU) alprostadil, intracavernosal injection (ICI) and prosthesis.
- The older **step-care / Process of Care Model (1999)** ranked them: oral, then local/VED, then prosthesis.
- In testosterone (T) deficiency (symptoms and total T below 300 ng/dL), restoring T probably improves the other ED treatments.
- A **partner interview** changes management in over half in some series and drives adherence. **Follow-up** reduces the high discontinuation rate.

Cardiac risk assessment – Princeton Consensus

- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *Princeton high (defer sex and ED treatment) / indeterminate (stress test, restratify) / low (treat)*

Not in the paper answer:

- Every risk group gets lifestyle counselling and risk-factor control.

Every man with ED, even asymptomatic, is at potential CV risk; sex equals moderate exertion. A Framingham or global risk score is a useful first step.

Refer after failed therapy, for a young trauma patient, for deformity, in medicolegal cases, and for complex endocrine (e.g. pituitary adenoma), psychiatric or vascular problems.

3 · Basic evaluation

Why this matters: simple tests uncover the disease behind the ED.

History

- Ask about onset, situational vs global pattern, morning/nocturnal and masturbatory erections, libido, ejaculation, curvature, partner and prior treatment. Is ED secondary to another phase, e.g. premature ejaculation?
- Ask about DM, CVD, neurological disease, pelvic surgery or radiation, trauma and drugs. Depression is bidirectional; use the Whooley two-question screen.

Psychogenic	Organic
Sudden onset	Gradual onset
Complete, immediate loss	Incremental progression
Situational	Global
Waking/nocturnal erections present	Poor or absent

> **KEY POINT**
> Preserved nocturnal/morning erections prove an intact neurovascular axis, the strongest historical pointer to psychogenic ED.

Questionnaires

- The **International Index of Erectile Function (IIEF)** has 15 items in 5 domains: erectile function, orgasm, desire, intercourse satisfaction and overall satisfaction.
- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *IIEF-5/SHIM bands: 5–7 severe … 22–25 no ED*
- The Sexual Health Inventory for Men (SHIM) has 5 items scored 1–5, covering the past 6 months †. Questionnaires grade severity and response but do **not** identify the cause.

Physical examination

- Hypogonadism: testes, gynaecomastia, virilisation. Vascular: blood pressure (BP), obesity, pulses.
- Neurogenic: genital sensation, bulbocavernosus reflex. Structural: plaque, micropenis, chordee.

Laboratory tests

- **Recommended**: fasting glucose or glycated haemoglobin (HbA1c), lipids, serum chemistry, full blood count and **morning (7–11 am) total testosterone**.
- Repeat a low T, because about one-third of mildly low values normalise, and add luteinising hormone (LH) and prolactin. Thyroid tests are optional; check prostate-specific antigen (PSA) if prostate disease is suspected or before T therapy.

> **KEY POINT**
> The basic work-up finds the disease behind the ED (undiagnosed DM, dyslipidaemia, hypogonadism, CV risk); it does not prove ED.

4 · Hormonal evaluation

Why this matters: low T and high prolactin are treatable causes.

- P4 Male menopause (andropause, late-onset hypogonadism) — *bioavailable T = free + albumin-bound; SHBG rises with age; LH separates primary/secondary; prolactin if LH low; pituitary MRI if T <150 ng/dL*

Not in the paper answer:

- Commercial free-T assays are unreliable; use calculated bioavailable T.
- MRI is also needed for persistent hyperprolactinaemia or mass effect.
- Prolactin suppresses gonadotrophin-releasing hormone (GnRH) pulses, lowering T and libido.
- Free T is 0.5–3%, sex hormone–binding globulin (SHBG)-bound ~30% and albumin-bound ~67%.
- SHBG is lowered by obesity, nephrotic syndrome, hypothyroidism and glucocorticoids, and raised by ageing, cirrhosis, hyperthyroidism, HIV, anticonvulsants and oestrogens. In DM, use bioavailable T.
- The AUA threshold is **300 ng/dL (10.4 nmol/L)**; above about 230 ng/dL, sexual function correlates only weakly with T. Order a pituitary magnetic resonance imaging (MRI) scan if T is **<150 ng/dL**.
- Hyperprolactinaemia causes: antipsychotics, tricyclics, opiates, prolactinoma, hypothyroidism, renal failure. Measure oestradiol for gynaecomastia.

5 · Specialised testing

Why this matters: surgical candidates need the vascular lesion defined.

For young trauma patients, primary ED, surgical candidates, medicolegal cases and failed therapy.

Test	Line	Answers	Indication
Combined intracavernosal injection and stimulation (CIS)	1st	Can the vessels produce a rigid erection?	Screening
Penile duplex Doppler	2nd	Arterial or veno-occlusive? Asymmetry?	Most reliable, least invasive
DICC (dynamic infusion cavernosometry/cavernosography)	3rd	Is there a venous leak, and where?	Trauma, lifelong ED, before venous surgery
Selective internal pudendal arteriography	3rd	Site of arterial lesion; is the inferior epigastric donor usable?	Young trauma patient before revascularisation

CIS test

- Inject alprostadil 10–20 µg or Bimix/Trimix 0.3 mL (27–29 G needle) into the lateral base; compress 5 min, stimulate, rate rigidity.
- A rigid, sustained erection means normal haemodynamics. Anxiety or a low dose gives false negatives (redose); up to a fifth with borderline inflow test falsely normal.
- If there is no detumescence by about 1 h, inject diluted phenylephrine (500 µg/mL) every 3–5 min.

Penile duplex Doppler

- Measure cavernous artery peak systolic velocity (PSV) and end-diastolic velocity (EDV) at the base every 5 min to 20 min after ICI; redose if poor.

Parameter	Arterial insufficiency	Veno-occlusive dysfunction
PSV	**<25 cm/s** (25–35 borderline; >35 normal)	Stays >25 cm/s
EDV	—	**>5 cm/s**
Resistive index (RI) = (PSV − EDV)/PSV	—	**<0.75** (normal >0.9)
Other	Acceleration time >122 ms; side-to-side PSV difference >10 cm/s; artery dilation <75%	Rapid detumescence
When assessed	Within 5 min	15–20 min (needs full relaxation)

KEY POINT

Venous leak can be diagnosed only with adequate inflow and full smooth muscle relaxation; an anxious patient mimics veno-occlusive dysfunction, so redose first. Anatomical variants (dorsal–cavernous communications, early branching) can falsely suggest arteriogenic ED.

DICC and arteriography

- DICC: after a vasodilator, saline infusion measures flow-to-maintain, pressure decay and cavernous artery systolic occlusion pressure (CASOP); cavernosography localises leaks. Veno-occlusive dysfunction means mean systolic pressure cannot be reached, or pressure falls fast after infusion stops.
- Penile–brachial index and plethysmography are obsolete.

Nocturnal penile tumescence and rigidity (NPTR)

- Sleep-lab NPTR with polysomnography is the standard.
- The home **RigiScan** uses base and coronal loops to record tumescence and **radial** rigidity. Normal is 4–5 erections a night, ≥ †60% rigid for ≥10 min. It cannot confirm rapid eye movement (REM) sleep, and radial is not axial rigidity.
- Use it for obscure cause, medicolegal cases and suspected psychogenic ED. It is falsely normal in multiple sclerosis, falsely abnormal in ageing, depression and anxiety.

Neurological tests (bulbocavernosus reflex latency, dorsal nerve conduction, biothesiometry) have a limited role. In DM, penile neuropathy can exist without foot neuropathy.

6 · Treatment

Why this matters: most men can be helped, and surgery is kept for failures.

ALGORITHM

1. Treat reversible causes (lifestyle, drugs, hormones, psychosocial) and stratify CV risk.
2. Shared decision: PDE5I, VED, IU alprostadil or ICI. Low-intensity extracorporeal shock wave therapy (Li-ESWT) is investigational.
3. If the response is inadequate, then optimise, combine or switch.
4. If still inadequate, then offer a penile prosthesis *(see Ch 75)*.

Lifestyle and risk-factor modification

- Stopping smoking, exercise, weight loss and a Mediterranean diet improve function; control DM, lipids and BP.

- **Change medication**: replace thiazides or β-blockers with calcium channel blockers, angiotensin-converting enzyme inhibitors (ACEIs) or angiotensin receptor blockers (ARBs). For a selective serotonin reuptake inhibitor (SSRI), switch to bupropion, mirtazapine or buspirone, take a drug holiday, or add a PDE5I.
- **Psychosexual therapy** (sensate focus, cognitive-behavioural therapy, couples therapy) works best with medical therapy.

Hormonal therapy

- Give T only for confirmed deficiency, as a ~3-month trial; it improves libido more than erection and is synergistic with PDE5Is.
- Monitor digital rectal examination, PSA, haematocrit, liver tests and lipids at baseline, 3–6 months and yearly. Adverse effects (AEs): erythrocytosis, sleep apnoea, acne, gynaecomastia, lower urinary tract symptoms. A meta-analysis of randomised controlled trials (RCTs) found no causal CV link.
- Preparations: gels (transfer risk), patches, intramuscular enanthate/cypionate (200–250 mg every 2–3 weeks; supraphysiological peaks, polycythaemia), undecanoate depot (750 mg at 0 and 4 weeks, then every 10 weeks), pellets, nasal gel (no transfer) and oral undecanoate (raises BP). Avoid 17α-alkylated methyltestosterone because it is hepatotoxic.
- T does not correct hyperprolactinaemia: stop the drug, give bromocriptine for prolactinoma, and operate for failure or visual compromise.

PDE5 inhibitors (first-line oral)

They block cyclic guanosine monophosphate (cGMP) breakdown, so they **enhance, not initiate**, erection; sexual stimulation is obligatory.

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Doses (mg)	25, 50, 100	5, 10, 20	5, 10, 20 (2.5/5 daily)	50, 100, 200
Time to peak, T_{max} (h)	0.8 (~1 h before)	0.7–0.9 (~1 h)	**2** (~2 h)	**0.3–0.5** (~30 min)
Onset (min)	15–60	15–60	15–120	15–60
Half-life (h)	3–5	4–5	**17.5**	3–5
Fatty food	Lowers absorption	Lowers absorption	**No effect**	Lowers absorption
Characteristic AE	Visual (blue tinge; PDE6), flushing	Visual, flushing; **QT prolongation, so avoid with class 1A/3 antiarrhythmics** and congenital long QT	**Back pain/myalgia** (PDE11)	Fewest visual effects
Nitrates	**Contraindicated (all)**			

- Common AEs: headache (commonest), dyspepsia, flushing, nasal congestion. Efficacy is equal: about 70% achieve intercourse, 40–50% in DM and after prostatectomy.
- **Nitrates** in any form, including amyl nitrite "poppers", cause profound hypotension; use more than 2 weeks earlier is not a contraindication. If angina occurs, give no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil).
- No antidote exists; treat hypotension with head-down tilt, IV fluids, phenylephrine, then an intra-aortic balloon pump.

- Caution: α-blockers (stabilise first, start low), aortic stenosis or outflow obstruction, hypovolaemia. CYP3A4 inhibitors (ketoconazole, ritonavir) need a lower dose; inducers (rifampicin) a higher one.
- Not studied: MI or stroke within 6 months, BP <90/50 or >170/100, retinitis pigmentosa, Child-Pugh C, dialysis.
- Nonarteritic ischaemic optic neuropathy (NAION) is not proven to increase; for sudden visual loss, stop and seek care.
- **Before labelling failure**, check stimulation, timing, food, maximum dose and **up to 9–10 attempts**; correct hypogonadism, glycaemia and lipids; then switch agent or try daily tadalafil.
- **Penile rehabilitation** after prostatectomy is unproven: a nightly sildenafil trial was positive, but REINVENT and the nightly-vs-on-demand trials were negative.

Other oral/central agents

Yohimbine (α2 antagonist) helps psychogenic ED only; sublingual apomorphine (D1/D2 agonist) causes nausea and is not US-approved; trazodone equals placebo; phentolamine helps minimal ED only.

Vacuum erection device

- Negative pressure fills the corpora and a base ring traps the blood; **remove it within 30 min**. The erection is venous, hinged at the base, cool and cyanotic.
- Up to 90% get an erection but only 30–70% are satisfied. It suits older men in stable relationships and helps a soft glans.
- AEs: pain, numbness, trapped ejaculate, petechiae, rarely skin necrosis. Take care with anticoagulants.

Intraurethral alprostadil (MUSE)

- A prostaglandin E1 (PGE1) pellet, 125–1000 µg, goes 3 cm into the urethra after voiding; then massage and stay upright.
- About 50% respond, so it is niche: PDE5I failure after nerve injury, PDE5I add-on, or soft glans after prosthesis. A constriction band (ACTIS) helps.
- AEs: urethral/penile pain (about one-third), minor bleeding, hypotension; priapism is rare. Use a condom if the partner is pregnant.

Intracavernosal injection

ICI acts directly on smooth muscle, **independent of nerves and nitric oxide (NO)**, so it works when PDE5Is fail (neurogenic, post-prostatectomy, DM).

Agent	Mechanism	Notes
Alprostadil (PGE1) 5–40 µg	Adenylate cyclase raises cyclic AMP (cAMP)	Only FDA-approved ICI; ~70% efficacy; 96% metabolised locally within 60 min; **penile pain** commonest AE; least priapism and fibrosis
Papaverine	Nonspecific PDE inhibitor; Ca^{2+} channel block	Cheap; raises liver enzymes, causes priapism and fibrosis, so not used alone
Phentolamine	α-antagonist	Adjunct only; hypotension, reflex tachycardia

Agent	Mechanism	Notes
Trimix (alprostadil 10 µg, papaverine 30 mg and phentolamine 1 mg per mL)	Synergistic	≈90%; rescues monotherapy failures; less pain
Bimix	Papaverine with phentolamine, or alprostadil with phentolamine	≈90%

- Titrate in the office from a low dose, since neurogenic and psychogenic ED are very sensitive. Inject the lateral shaft (2 or 10 o'clock †), avoiding the dorsal bundle and urethra. Aim for an erection of 1 h or less.
- **Priapism** is highest with papaverine mixes and in neurogenic or young men. Seek care beyond †4 h for aspiration with or without phenylephrine. Long-term use causes fibrosis and curvature.
- Contraindications: psychological instability, priapism risk (sickle cell), severe coagulopathy, unstable CVD, poor dexterity (partner may inject), and monoamine oxidase (MAO) inhibitors (phenylephrine reversal risks hypertensive crisis).

Topical agents

Alprostadil cream (Vitaros) is modest. MD3000 (Eroxon) is the first over-the-counter FDA-cleared glans gel.

Restorative and investigational

Therapy	Proposed mechanism	Evidence/status
Li-ESWT	New vessels; nerve, endothelial, smooth muscle regeneration	RCT meta-analysis: IIEF better than sham in **mild–moderate vasculogenic** ED, the best evidence; still investigational (Sexual Medicine Society of North America: research only)
Platelet-rich plasma	Growth factors	Conflicting RCTs
Stem cells	Regeneration	Uncontrolled series only
Botulinum toxin ICI	Blocks sympathetic noradrenaline	Short-term benefit in refractory ED

Combination therapy

Monotherapy fails in nearly 40%. Pairs: daily tadalafil 5 mg with on-demand sildenafil; PDE5I with T; PDE5I with IU, ICI or VED; ICI with VED; prosthesis with IU for a soft glans. PDE5I with ICI is off-label, so test it in the office.

Surgery

- A **penile prosthesis** suits men whose other therapy is contraindicated, ineffective or unwanted, corporal fibrosis after ischaemic priapism or infection, and in Peyronie disease *(see Ch 75)*.
- P2 Surgical management of erectile dysfunction and penile prosthesis — *arterial revascularisation (age <55, nonsmoker, nondiabetic, focal post-traumatic pudendal stenosis; glans hyperaemia) and investigational venous ligation*

Not in the paper answer:

- Venous ligation is investigational because a leak is usually a smooth muscle disorder.

7 · Landmark trials

Why this matters: these trials shape CV work-up and rehabilitation.

Trial	Design	Result	Take-home
Prostate Cancer Prevention Trial (placebo arm)	Prospective cohort, men ≥55	ED gave 45% more cardiac events at 5 years	ED is a CV sentinel; stratify risk
REINVENT	RCT: nightly vs on-demand vardenafil vs placebo after nerve-sparing prostatectomy	No difference in spontaneous recovery	Scheduled "rehabilitation" PDE5I is unproven

73 Priapism

1 · Definition and classification

Why this matters: the type decides urgency; ischaemic priapism destroys tissue within hours.

- **Priapism** is a full or partial erection lasting **≥4 h** beyond, or unrelated to, sexual stimulation. It is an emergency because it causes fibrosis, shortening and permanent erectile dysfunction (ED).
- It is classified by **oxygenation, not duration**: failed outflow causes time-critical ischaemia, while excess arterial inflow gives an oxygenated, painless erection that can wait.

Feature	Ischaemic (low-flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high-flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Failed outflow, like a compartment syndrome	Repeated self-limiting episodes, often nocturnal	Fistula between cavernosal artery and sinusoids
Typical cause	Drugs, sickle cell disease (SCD), blood disorders	SCD	Blunt perineal/penile trauma, needle injury
Pain / rigidity	Painful (within 6–8 h); fully rigid, tender	Painful; rigid during episode	Painless; partially rigid
Blood gas / Doppler	Low oxygen, high CO_2, acidotic; absent or low flow	Ischaemic in each episode	Arterial gas; high flow, fistula or blush
Management	Emergency decompression	Treat episode as ischaemic, plus prophylaxis	Observe, then embolise
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

> **KEY POINT**
> Trap: stuttering priapism is **ischaemic**, so any episode lasting ≥4 h needs the same first-line treatment.

2 · Aetiology

Why this matters: the cause guides treatment and prevention.

- P3 Priapism: Diagnosis and Management — *causes: ICI, α-blockers, trazodone, antipsychotics, cocaine, anticoagulant withdrawal, SCD, leukaemia, metastases, venom; non-ischaemic trauma with delayed onset*

Not in the paper answer:

- Most drugs block α-mediated detumescence or relax smooth muscle directly.
- Metastases (testis, kidney, lung) obstruct venous outflow.

Ischaemic – extra points

- Intracavernosal injection (ICI) misuse or recreational use: younger men, late presentation, often HIV-positive.
- Other drugs: hydralazine, propranolol, oral phosphodiesterase type 5 inhibitors (PDE5Is), attention-deficit drugs (methylphenidate, atomoxetine), gonadotrophin-releasing hormone (GnRH), testosterone, hydroxyzine, alcohol and marijuana. Take a full drug history.
- Blood causes: thalassaemia, myeloma, asplenia, erythropoietin, dialysis with heparin. In leukaemia, give cytotoxics and leukapheresis alongside penile treatment.

- **Recurrence after a shunt** may mean shunt thrombosis. Give **aspirin 325 mg + heparin 5000 U subcutaneously** before surgery, then aspirin 81 mg daily for 2 weeks.

Sickle cell disease

- Haemoglobin S (HbS) polymerises when deoxygenated, causing haemolysis and low nitric oxide (NO), which obstruct outflow.
- Lifetime risk is **29–42%**; SCD causes 23% of adult and **63% of paediatric** ischaemic priapism. Onset is about 15 years; stuttering is commoner than major episodes.
- Triggers: arousal, sleep, fever, cold, dehydration. Sickle trait: low risk.

Non-ischaemic

- Tumour erosion is another cause, so think of malignancy over 55 years. Onset is often delayed ≥1 day, because the clot over the tear disperses in a nocturnal erection.
- Transient **post-ischaemic hyperaemia** after treatment mimics it; colour Doppler distinguishes.

3 · Molecular pathophysiology

Why this matters: priapism is failed detumescence, which is why a daily PDE5I prevents it.

- Low basal NO from endothelial nitric oxide synthase (eNOS) reduces cyclic guanosine monophosphate (cGMP) feedback, so **PDE5 is down-regulated**. A neural NO surge then gives cGMP that cannot be broken down, so relaxation is unopposed.
- Rho-kinase is also lowered and adenosine raised (via HIF-1α).
- Prolonged hypoxia causes smooth muscle apoptosis and transforming growth factor-β (TGF-β) fibrosis, leading to ED.
- Paradox: **a daily PDE5I restores PDE5 and prevents** recurrent episodes.

4 · Evaluation

Why this matters: a corporal blood gas sorts the type in minutes.

- P3 P1 Priapism: Diagnosis and Management; Doppler in urology — *history, examination, corporal blood gas table (ischaemic pO_2 <30, pCO_2 >60, pH <7.25; flaccid 40/50/7.35; arterial >90/<40/7.4), PDUS (PSV <50 vs >50 cm/s), parallel labs*

Not in the paper answer:

- Record baseline erectile function; it is needed for prosthesis counselling.
- Penile duplex ultrasound (PDUS) is most useful for recurrence after treatment.
- SCD labs: low haemoglobin; high reticulocytes, lactate dehydrogenase, bilirubin.
- Glans and spongiosum stay soft in both types; involvement suggests another disease. Check vitals before sympathomimetics.
- Gas may be skipped after known recent ICI. Check colour first: dark red (ischaemic) or bright red. Magnetic resonance imaging (MRI) shows viability in non-acute cases.

ALGORITHM

1. History, examination and corporal blood gas.
2. If ischaemic, then acute ischaemic algorithm.
3. If indeterminate, then penile/perineal PDUS.
4. If non-ischaemic, then expectant algorithm.

5 · Acute ischaemic priapism (AUA/SMSNA)

Why this matters: every hour counts, so decompress and escalate fast.

- Irreversible damage begins 4–12 h after presentation; counsel about ED risk first. Cold packs, exercise, ejaculation, oral drugs and IV fluids are **adjuncts only; never delay first-line treatment**. Oral terbutaline/pseudoephedrine is not recommended.
- P3 Priapism: Diagnosis and Management — *ED by duration (SCD series): ≤12 h 0%, 12–24 h 22%, 24–36 h 56%, ≥36 h **100%***

ALGORITHM

1. If <36 h, then aspirate ± irrigate and inject intracavernosal phenylephrine (BP and cardiac monitoring).
2. If that fails, then distal shunt ± tunnelling; tunnel if not done.
3. If still erect, then proximal shunt or penoscrotal decompression; if refractory, prosthesis.
4. If >36 h, then counsel high ED risk with analgesia/follow-up, OR early penile prosthesis.

Aspiration / irrigation

- After a penile block, insert a **19-G needle** at the penoscrotal junction, **3 or 9 o'clock**. †One side usually suffices (septum fenestrated). Aspirate with shaft compression, repeat, irrigate saline.
- Aspiration alone works in fewer than 1 in 4. Cold (10 °C) irrigation gave 96% vs 60% in a randomised controlled trial (RCT); **avoid chilled saline in SCD**.

Phenylephrine

- A selective α1-agonist with minimal β action, so least cardiac effect.
- Dilute to **100–500 µg/mL**; give **1-mL aliquots every 3–5 min** for up to **1 h** before failure; †maximum ≈1 mg in 1 h (European Association of Urology, EAU).
- Toxicity: hypertension, headache, arrhythmia, **reflex bradycardia**. Monitor, especially with cardiovascular disease (CVD) or monoamine oxidase (MAO) inhibitors.

Shunts

- P3 P2 Priapism: Diagnosis and Management; Management of priapism — *Winter, Ebbehoj, T-shunt, Al-Ghorab, snake tunnelling (7/8 Hegar), Quackels/Sacher (stagger ≥1 cm), Grayhack/Barry, penoscrotal decompression*

Not in the paper answer:

- Ebbehoj is a bedside No. 11 scalpel stab; Al-Ghorab needs sedation, spinal or general anaesthesia but is durable.
- Distal shunts make a cavernosum–spongiosum fistula that closes after detumescence; none are compared head to head. Winter is bedside, least invasive, least successful. T-shunt: No. 10 blade lateral to meatus, bilateral if needed. Al-Ghorab: transverse incision 1 cm distal to dorsal corona. Snake rescues prolonged or failed cases.
- Quackels is unilateral and Sacher bilateral (1-cm ellipses, 5-0 polydioxanone (PDS)). Grayhack mobilises ~10 cm of saphenous vein. Morey penoscrotal decompression (paediatric Yankauer) spares the glans and permits a concurrent prosthesis.
- AUA: proximal shunts only after failed distal shunt with tunnelling. Complications: oedema, haematoma, urethral injury or fistula, necrosis, infection and pulmonary embolism (PE).

Early penile prosthesis

- For priapism **>36 h** or failed shunts; oedematous corpora dilate easily, while delayed surgery meets dense fibrosis.
- Early vs delayed: shortening 3% vs 40%, fewer revisions and infections, satisfaction 96% vs 60%; rigidity is similar.
- Prior distal shunts raise distal perforation risk; if recovery is unlikely, weigh shunting against a future implant *(see Ch 75)*.

6 · Non-ischaemic priapism

*Why this matters: it is **not an emergency**.*

- About 60% resolve with observation, compression and ice.

ALGORITHM

1. Observe and reassess at 4 weeks.
2. If it persists, then PDUS (fistula site, size), counsel, and selectively embolise the internal pudendal/cavernosal artery.
3. If it recurs, then repeat embolisation with a non-resorbable agent.
4. Only after repeated failure, ligate surgically.

Agent	Type	Feature
Autologous clot	Temporary	Children; early recurrence
Gelatin sponge	Temporary (2–6 weeks)	Reabsorbed
Microcoils	Permanent	Theoretical ED risk
Polyvinyl alcohol particles	Permanent	**Preferred for repeat** (AUA)
N-butyl cyanoacrylate	Permanent	Second line for recurrence

- Embolisation success ≥75%, recurrence 30–40%. Open ligation: about 50% ED, gangrene, abscess.

7 · Stuttering priapism — prevention

Why this matters: each episode adds damage.

- An episode ≥4 h is acute ischaemic priapism. Short episodes: fluids, walking, analgesia; home self-injected phenylephrine for breakthrough.
- No prophylactic drug has robust evidence.

Agent	Dose	Note
PDE5I	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Most promising; †start when flaccid; the only RCT missed its primary endpoint
Pseudoephedrine	60–120 mg at night	Off-label first-line α-agonist
Etilefrine	50–100 mg daily	α1-agonist; Cochrane: insufficient evidence
GnRH agonists, antiandrogens (bicalutamide, flutamide, cyproterone)	Depot / oral	Suppress libido and erections; †avoid before puberty is complete
Finasteride	5 mg, tapered to 3 mg then 1 mg	One SCD series

Agent	Dose	Note
Digoxin	0.25–0.5 mg	Inhibits Na^+/K^+-ATPase
Baclofen	Oral 40 mg at night; intrathecal	Neurogenic/reflex erections

SCD specifics

- Same first-line aspiration and phenylephrine; IV hydration and analgesia **concurrently, never instead**. †Exchange transfusion is not a primary treatment (AUA); involve haematology. Avoid chilled irrigation.

8 · Landmark trials

Why this matters: evidence for irrigation, prophylaxis and early implants.

Trial	Design	Result	Take-home
Ateyah 2005	RCT, ICI-induced priapism, irrigation temperature	10 °C saline 96% vs 60% at 37 °C	Cold irrigation (not in SCD)
Burnett 2014	RCT, SCD, sildenafil vs placebo	Primary endpoint missed; open-label benefit	PDE5I prophylaxis plausible, unproven
Zacharakis 2014	Early vs delayed prosthesis	Shortening 3% vs 40%; satisfaction 96% vs 60%	Implant early when ED is inevitable

9 · MCQ one-liners

MCQ ONE-LINERS

1. The first case report of priapism was in 1824.

74 Disorders of Male Orgasm and Ejaculation

1 · Spectrum

Why this matters: these are the commonest male sexual complaints and a cause of infertility.

- Ejaculatory dysfunction (EjD) is the **most common male sexual disorder**. It spans premature ejaculation (PE), delayed ejaculation (DE) and anejaculation, plus retrograde ejaculation (RE), painful ejaculation, anhedonia and postorgasmic illness syndrome (POIS).
- Orgasm is a separate **cortical** event, so ejaculation and orgasm can dissociate.

2 · Physiology of ejaculation

Why this matters: the nerves and transmitters of each phase explain causes and drugs.

- Lumbar interneurons at **L3–L5** form the **spinal ejaculation generator**. They coordinate thoracolumbar and lumbosacral nuclei under genital and brain input.
- Brain: bed nucleus of the stria terminalis, medial amygdala, posterodorsal preoptic nucleus, parvicellular subparafascicular thalamus, medial preoptic area (MPOA) and paraventricular nucleus (PVN).

Phase	Events	Mediation
Emission	Vas, seminal vesicles and prostate deliver semen to the posterior urethra	Autonomic, **T10–L2**
Ejection	Bulbospongiosus, ischiocavernosus, levator ani contract rhythmically; **bladder neck closes**; external sphincter relaxes	Somatic **S2–S4 (pudendal)** and a sympathetic reflex
Orgasm	Cerebral processing of pudendal input	Cortical

- **Dopamine (D2) promotes** and **serotonin (5-HT) inhibits**; noradrenaline, acetylcholine, oxytocin, GABA and nitric oxide also act.
- 5-HT2C stimulation delays and 5-HT1A shortens ejaculation, so lifelong PE is thought to be **5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity**.

> **KEY POINT**
> Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic.

3 · Premature ejaculation — definition and classification

Why this matters: short latency alone is not PE; loss of control and distress are needed too.

- P1 P3 Premature Ejaculation — *ISSM 2014 three criteria; lifelong/acquired/variable/subjective; AUA 2 min; aetiology, evaluation (IELT, PEDT, IIEF-5), drug table*

Not in the paper answer:

- DSM-5 duration (≥6 months, 75–100% of occasions) and severity bands (table below).
- SSRI cautions beyond bipolar disorder: under-18s, bleeding with NSAIDs, sperm DNA damage (Section 5).

	Lifelong	Acquired	Variable	Subjective
Onset	From first encounters, every partner	After previously normal latency	—	—

	Lifelong	Acquired	Variable	Subjective
Latency	≤ ~1 min (mostly 30–60 s)	Markedly reduced, often ≤3 min	Occasionally short	Normal or long
Basis	Neurobiological, genetic	Medical/psychological	Normal variation	Psychological/cultural
Disorder?	Yes	Yes	No	No

- Waldinger subtypes: variable and subjective PE dominate the community, lifelong and acquired the clinic.
- The International Society for Sexual Medicine (ISSM) requires ejaculation within **about 1 min** (lifelong) or a bothersome fall, often to **≤ about 3 min** (acquired). It is the first **evidence-based** definition, for vaginal intercourse only; "about 1 min" allows ~90 s. **Latency alone never suffices**.

	ISSM	DSM-5 (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ ~1 min	≈1 min	≈**2 min** since sexual debut
Acquired	Bothersome fall, often ≤3 min	Specifier only	Marked fall (2–3 min or ≥50%)
Duration/frequency	Always or nearly always	≥6 months, 75–100% of occasions	—
Severity	—	Mild 30–60 s; moderate 15–30 s; severe <15 s	—
Basis	Evidence	Same data	Expert opinion

- DSM-5: Diagnostic and Statistical Manual of Mental Disorders; AUA/SMSNA: American Urological Association/Sexual Medicine Society of North America. The AUA 2-min line trades specificity for sensitivity.
- MCQ ICD-10 is the only older definition with a number (≤15 s); Masters and Johnson: partner satisfied in <50% of episodes.
- Community stopwatch intravaginal ejaculatory latency time (IELT) is positively skewed, median **5.4 min**; ≤1 min is statistically abnormal.

4 · Premature ejaculation — aetiology and evaluation

Why this matters: acquired PE usually has a treatable cause.

- Lifelong PE: younger men, fewer comorbidities. Acquired PE: older men, metabolic syndrome, diabetes.
- **PE with erectile dysfunction (ED)** affects up to half of ED men, who "rush" before a partial erection is lost. Pitfall: PE men mistake post-ejaculatory detumescence for ED, giving a false-positive Sexual Health Inventory for Men (SHIM).
- In **chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS)**, PE is the main sexual symptom; antibiotics for proven bacterial prostatitis lengthen IELT.
- In **hyperthyroidism**, PE tracks suppressed thyroid-stimulating hormone (TSH) and improves when euthyroid, but routine TSH is not indicated. Circumcision has no effect.
- Normal International Index of Erectile Function (IIEF) erectile domain is ≥26 (IIEF-5 ≥22). Examination is desirable in lifelong and **mandatory in acquired PE**, with digital rectal examination (DRE) for prostatitis.

5 · Premature ejaculation — treatment

Why this matters: drugs work, but most men stop them.

ALGORITHM

1. Confirm PE.
2. If it is secondary to ED or another dysfunction, then treat that primary cause.
3. If lifelong, then SSRI ± behavioural therapy (patient preference).
4. If acquired, then treat the cause ± behavioural therapy/SSRI; try withdrawal after 6–8 weeks.
5. If variable or subjective, then reassurance, education, psychotherapy.

- Everyone gets psychosexual education. Cognitive-behavioural therapy is limited first-line in lifelong PE but an important adjunct in acquired PE.

Drug	Dose / timing	Notes
Dapoxetine	30–60 mg, 1–3 h before	**Only licensed oral PE drug** (>50 countries, not FDA); short half-life; works from the first dose; no PDE5I interaction; many stop it
Paroxetine (daily)	10–40 mg	**Strongest delay**; off-label
Sertraline / fluoxetine / citalopram	50–200 / 20–40 / 20–40 mg daily	Off-label
Clomipramine (a tricyclic antidepressant, TCA)	12.5–50 mg daily or 3–4 h before	Off-label
Topical lidocaine/prilocaine	20–30 min before (spray ≥5 min)	Condom advised; level 1a evidence
Phosphodiesterase type 5 inhibitors (PDE5Is)	30–50 min before	Best with coexisting ED; add-on to SSRI
Tramadol	25–50 mg, 3–4 h before	Low evidence; addiction risk

- **Daily selective serotonin reuptake inhibitors (SSRIs)** act in 5–10 days, fully at **2–3 weeks**. Early side effects: fatigue, yawning, nausea, diarrhoea, sweating.
- Cautions: bipolar disorder (hypomania), bleeding with aspirin/NSAIDs, under-18s (suicidality), sperm DNA fragmentation. **Never stop them abruptly.** Off-label SSRIs taken on demand delay less.
- Topicals numb the penis, and the partner without a condom; the metered spray (Fortacin) does not penetrate keratinised skin, so transfer is minimal.
- Meta-analyses: PDE5I beats placebo and PDE5I with SSRI beats SSRI alone, so PDE5Is are allowed with or without ED.

On-demand dapoxetine	Daily off-label SSRI
Licensed; fewer side effects; little withdrawal; infrequent intercourse	Greater delay; full effect 2–3 weeks; must taper; established relationships

KEY POINT

The real limiter in PE is adherence, not efficacy: most stop SSRIs and topicals (cost, lost spontaneity, "antidepressant" stigma).

6 · Delayed ejaculation, anejaculation and anorgasmia

Why this matters: one reflex arc gives DE, anejaculation, RE and anorgasmia; history finds the level.

- DE (DSM-5): marked delay or absence despite adequate stimulation, ~75–100% of occasions, with distress. There is **no validated cut-off**; most men ejaculate at 4–10 min, so >25–30 min with distress suggests DE.
- **Anorgasmia** is absent perceived orgasm. DE is ~25% lifelong, ~75% acquired, and rises with age. Hallmark: normal erections, **low subjective arousal**.

Category	Causes
Afferent	Ageing (penile afferents, Pacinian corpuscles)
Psychogenic	Inhibited ejaculation; performance anxiety; religious orthodoxy; **idiosyncratic vigorous masturbation**
Endocrine	Hypogonadism, **hypothyroidism** (DE; hyperthyroidism causes PE), hyperprolactinaemia
Neurogenic disease	Spinal cord injury (SCI), then diabetic autonomic neuropathy, then multiple sclerosis (MS)
Neurogenic surgery	Radical prostatectomy, rectal surgery, sympathectomy, aortic aneurysm repair, retroperitoneal lymph node dissection (RPLND)
Anatomical (causes RE)	Transurethral resection of prostate (TURP), bladder neck incision
Congenital / infective	Müllerian cyst, Wolffian anomaly, prune belly; urethritis, tuberculosis (TB), schistosomiasis
Drugs	**SSRIs** (~7-fold DE risk), TCAs, antipsychotics, α-methyldopa, thiazides, alcohol

- Antipsychotics cause DE (dopamine block, prolactin) and RE (α-block).
- **Superior hypogastric plexus** injury at RPLND causes dry ejaculation; **nerve-sparing RPLND**, like total mesorectal excision with autonomic nerve preservation, preserves it in most.
- After radical prostatectomy: no ejaculation, orgasm preserved, possible climacturia. After radiotherapy: lower volume, weaker orgasm.
- Ask why he stops, whether masturbation works (coital-specific vs global), and "ejaculated but no semen?" (RE or obstruction).
- Examine vasa, DRE, bulbocavernosus reflex; check morning testosterone (T), HbA1c and lipids.

ALGORITHM

1. Orgasm? If never or sometimes, then inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique).
2. If always but no ejaculate, then look for **sperm in post-orgasm urine**; if present, RE.
3. If absent (aspermia), then failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO).

	Bilateral EDO	RE	Failure of emission
Antegrade volume	Very low	Low or absent	Absent
pH, fructose	Low (seminal vesicle fluid, ~80% of volume, is missing)	—	—
Post-orgasm urine	No sperm	**Sperm and fructose**	No sperm

Treatment

- Treat the cause; counsel about anejaculation and sperm retrieval before pelvic surgery. Refer to a psychosexual therapist once organic/drug causes are excluded (AUA).
- Normalise T, change the offending drug, increase stimulation, change masturbatory style.
- Drugs are all off-label, ~40% success, and poor in psychogenic/neuropathic DE (usually irreversible).

Drug	Action	Dose
Cabergoline	Central D2 agonist	0.5–2 mg every 3 days; **annual echocardiography** (valvular fibrosis)
Bupropion	Dopaminergic	150 mg once or twice daily
Pseudoephedrine	α1 agonist	120 mg 1–2 h before
Reboxetine	Noradrenaline reuptake inhibitor	4–8 mg daily
Cyproheptadine	5-HT antagonist	4–12 mg 3–4 h before (SSRI-induced DE); sedation
Amantadine, buspirone	—	Anecdotal
Intranasal oxytocin	Oxytocinergic	Both RCTs negative

7 · Retrograde ejaculation

Why this matters: RE follows prostate surgery and diabetes, but urinary sperm can still be used.

- P3 Retrograde ejaculation — *bladder neck failure; TURP 42–100%, TUIP 5–45%, diabetic neuropathy; post-ejaculate urine; pseudoephedrine/imipramine doses; urinary sperm retrieval for IUI/IVF/ICSI*

Not in the paper answer:

- Transurethral incision (TUIP) causes less RE if one incision stops short of the verumontanum.
- Aquablation beats TURP for ejaculation in randomised controlled trials (RCTs).
- Antimuscarinic with sympathomimetic beats either alone.
- Diagnosis: **>5–10 sperm/high-power field** in urine passed after masturbation confirms RE.

8 · Sperm retrieval in SCI

Why this matters: most men with SCI cannot ejaculate, yet sperm can usually be obtained.

- Reflex erection needs the sacral arc (preserved in upper motor neuron (UMN) lesions); ejaculation needs thoracolumbar outflow, so it is rare (<5%) with complete UMN lesions. RE is common.

ALGORITHM

1. **Penile vibratory stimulation (PVS)** is first line, with up to ~70% success; it needs an intact reflex at the T12–S1 level.
2. If PVS fails, then **electroejaculation (EEJ)**: a rectal probe stimulates hypogastric sympathetic efferents; effective and safe.
3. If EEJ fails, then prostate massage, then surgical retrieval (TESE/TESA/MESA/PESA); testicular sperm extraction succeeds in most.

- For RE, empty the bladder and instil sperm-wash medium.
- Lesions above T6 risk **autonomic dysreflexia**: pretreat with **nifedipine**, monitor BP.
- Motility is low at first but improves with repeated ejaculation; fresh testicular sperm may beat frozen.

9 · Other disorders

Why this matters: rarer, distressing complaints missed unless asked.

Disorder	Key points
Painful ejaculation	Prostatitis/CPPS, urethritis, benign prostatic hyperplasia (BPH), seminal vesicle stones, EDO, α-blockers (least with alfuzosin); often none. Surgery (TURP, transurethral resection of ejaculatory ducts (TURED), pudendal neurolysis) weakly supported
Haematospermia	Usually benign (see pointer below). Seminal vesiculoscopy if refractory *(see Ch 70)*; †evaluate if >40 y or persistent
Climacturia	Leak at orgasm after radical prostatectomy. Steps: void first, condom, pelvic floor training, constriction ring, male sling
Ejaculatory anhedonia	Orgasm without pleasure; depression, hypogonadism, hyperprolactinaemia, SSRIs
Post-SSRI sexual dysfunction	Genital numbness, low desire, ED, anhedonia persisting months to years
POIS	Myalgia, fatigue, flu-like state within 30 min of orgasm
Orgasmic headache	Dull bilateral ache rising with excitement; sudden, intense at orgasm

- P1 Haematospermia — *causes, who needs work-up (≥40 y, persistent, haematuria), PSA/TRUS/MRI, treatment*

Not in the paper answer:

- Weight loss and bleeding disorder are further red flags.

75 Surgery for Erectile Dysfunction

1 · Place of surgery and devices

Why this matters: a prosthesis is the last step for erectile dysfunction (ED).

- P2 P3 Surgical management of erectile dysfunction and penile prosthesis; Types of penile prosthesis — *last-line but most effective (satisfaction 80–90%; repair 15% at 5 y, 30% at 10 y); Lue goal-directed approach; malleable vs 2-piece vs 3-piece; device names, expansion, coatings; consent; pre-op preparation; revascularisation*

Not in the paper answer:

- Extensive testing for the cause of ED does not change the outcome.
- Rods have fixed girths and are sized 0.5–1 cm shorter than corporal length so they bend easily.
- Coloplast cylinders are polyurethane (Bioflex), so keep cautery **<35 W**; Boston and Rigicon silicone–fabric–silicone cylinders tolerate cautery.
- Length-expanding cylinders correct curvature poorly; narrow cylinders suit small or scarred corpora.
- Rod mechanical failure is exceedingly rare. The two-piece device was never popular (Ambicor discontinued).

2 · Selection and counselling

Why this matters: consent and preparation prevent complaints and infection.

- **Consent is expectation management**: the device restores rigidity, not the former erection.
- The commonest disappointment is a **shorter erection**, because the pseudocapsule limits expansion (worse in Peyronie disease). Document it; patients may be litigious.
- With **phimosis**, circumcise and implant about 6 weeks later. Elective circumcision with redundant foreskin is done at the same sitting (implant first, then circumcise).
- Raised glucose on the day needs intravenous (IV) insulin. Nasal methicillin-resistant *Staphylococcus aureus* (MRSA) carriers get chlorhexidine bathing and mupirocin 2% for 5 days.

3 · Surgical approaches

Why this matters: each incision endangers a different structure.

Approach	Advantages	Cautions
Penoscrotal	Majority choice; secure subdartos pump; good crural exposure	Urethra at risk, so Foley first; reservoir via inguinal canal
Infrapubic (Scott)	Reservoir under vision	Dorsal neurovascular bundle: stay sutures at 10 and 2 o'clock; patient pulls pump down daily
Subcoronal	Rods; allows straightening	Circumcise at the same time, or lymphoedema follows
Ventral penile	Rods without bending; foreskin kept	—

4 · Operative technique (three-piece)

Why this matters: correct dilation and sizing prevent most errors.

ALGORITHM

1. Catheter; expose the tunica.
2. Two 2-0 polydioxanone (PDS) stay sutures per corpus; longitudinal corporotomy.
3. Dilate proximal, then distal (Metzenbaum, Hegar/Brooks 7→13 mm).
4. Goalpost test; measure (Furlow).
5. Cylinders in; test inflation 1. Close corporotomies; test inflation 2.
6. Reservoir, then pump; connect; test inflation 3.
7. Irrigate and close.

- Dilators point **dorsolaterally**, away from urethra and septum, and go gently to the ischial tuberosity. The **goalpost test** (equal height and symmetry) detects crossover.
- The cylinder is equal to or 1 cm shorter than measured; oversizing causes buckling and S-deformity. Rear tip extenders adjust length. Minimal dilation preserves length in primary implants.
- The pump sits in a dependent subdartos pouch, anterolateral to the testis. For malleable girth, two parallel Hegars slightly separated is ideal.

Reservoir placement

- **Retzius**: enter via the linea alba, or through the inguinal canal piercing transversalis fascia above the pubic tubercle. Empty the bladder; the iliac vessels lie ~1 inch lateral.
- In a hostile pelvis (radiation, cystectomy, robotic prostatectomy, transplant) use an **ectopic** site, made safe by the **lockout valve**.
- **Submuscular suprafascial** is the commonest ectopic site. Fill it nearly full, because underfilling leads to migration, and exclude intraperitoneal placement. Others: lateral retroperitoneal, epigastric, subcutaneous.
- Large reservoirs may compress the iliac vein, causing leg oedema.

5 · Intra-operative problems

Why this matters: each problem has a standard fix.

- **Fibrotic corpora** follow priapism or explant; pelvic radiation does *not* add difficulty. Use a midshaft corporotomy, cavernotomes (Rossello, Uramix), an Otis urethrotome or scar excision, and **narrow cylinders**, always two. Last resort is a graft (biologic oversized; synthetic sized to defect).
- **Crossover** gives a lopsided erection and unequal measurements. Leave a large dilator in the common cavity and redilate the other side laterally; the Foley should lie midline between cylinders.
- **Crural perforation** (one dilator deeper, lengths differ by >1 cm) is repaired with a **suture sling** (2-0 Prolene through the rear tip, over the corporotomy). Windsock is abandoned.
- **Urethral perforation** at the corporotomy: close it and proceed. Distal (fossa navicularis; irrigant at meatus): **abandon that cylinder**, place the contralateral one, leave no catheter and replace at 5–6 months.
- **Hypermobile glans** drooping ventrally is the supersonic transporter (**SST**) deformity, like a Concorde nose. Use glans fixation sutures (3-0 non-absorbable) via a hemicircumcising incision; distinguish it from short cylinders.
- **Residual curvature**: use high-rigidity cylinders (AMS 700 CX, Titan, Infla 10 X) and **modelling** (clamp tubing, bend 90° away for 90 s); then plication or incision with or without

graft. <20° is acceptable *(see Ch 76)*.

> **KEY POINT**
> No catheter after distal urethral perforation: the defect is not closed, and the urinary stream draws fluid into the urethra (Bernoulli), so urine does not enter it. Wait 5–6 months for scar to obliterate the false tract. Most intra-operative errors are **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley).

> **KEY POINT**
> After robotic prostatectomy the peritoneum is not replaced and bowel fixes near the internal ring, so a blind inguinal Retzius reservoir can compress bowel. Choose an ectopic site or place it under vision.

6 · Complications

Why this matters: infection usually costs the whole device.

Infection (1–3%)

- Risks are acute hyperglycaemia, revision (disturbed biofilm), skin or urinary sources, MRSA and scarred corpora.
- American Urological Association (AUA) prophylaxis: IV vancomycin, or 3rd-generation cephalosporin plus gentamicin, 1 h before incision and for ≤24 h. Extended oral courses: unproven. High risk: add an antifungal, with or without piperacillin–tazobactam.
- Coatings (InhibiZone rifampin–minocycline; hydrophilic Titan/Infla dipped) halve infection. **No-touch** technique gives <0.5%, because organisms enter via the wound.
- Fever and erythema suggest it; pus, exposed device, progressive pain or a **pump fixed to the scrotal wall** are definitive. **Remove ALL components**, as retained parts harbour biofilm.

Option	Technique	Caveats
Explant, delayed replacement	Antibiotic irrigation drains for 3 days; reimplant at 2–6 months	Fibrosis, loss of several cm; vacuum erection device (VED) meanwhile
Mulcahy salvage	Remove all, antiseptic washes, new drapes, immediate new device	~84%; fails with onset <2 months, aggressive organisms, purulence with cellulitis
Delayed salvage	IV vancomycin and gentamicin, salvage at 3 days	Aggressive infection
Malleable salvage	Rods instead	Quick; convert later

- Washes: dilute povidone-iodine (0.35–3.5%) for 3 min, then saline or antibiotic. **Hydrogen peroxide is dropped** (cytotoxic).
- **Revision washout**: an uninfected revision has ~10% infection, falling to ~3% with antiseptic washout of all cavities.

Mechanical failure

- This is the **commonest complication**: leak, pump malfunction, autoinflation, kinking or cylinder aneurysm.
- If the device is >3 years old, replace **all** parts; check any retained part with the **ohmmeter test** (saline conducts through a leak). A rod that loses rigidity has a core fracture and is replaced.

Extrusion and erosion

Feature	Extrusion	Erosion
Position	Tip through tunica, still under skin/glans	Through skin or into urethra
Cause	Forceful dilation, oversizing, aggressive use	Infection, crossover, catheterisation; rods in insensate men (spinal cord injury)
Management	**Distal corporoplasty** (new dorsomedial plane behind pseudocapsule)	Remove all, washout, contralateral spacer, replace at 5–6 months

- **Reservoir erosion** into bladder: exchange the reservoir and repair. Into bowel: remove the whole device.
- **Penile necrosis** risks are diabetes, vascular disease, smoking, prior implant and radiation. Avoid circumferential incisions, wrap with the glans visible, and remove cylinders promptly if it occurs.
- An early basal bulge is corporotomy disruption (repair); a late one is cylinder aneurysm (replace).

7 · Special situations

Why this matters: scarred penises need different timing or devices.

- After **priapism** >36 h, an **immediate implant** preserves length and relieves pain. Biopsy the necrosis, close distal shunt defects and avoid distally expanding cylinders (LGX, Infla 10 AX) *(see Ch 73)*.
- In **Peyronie disease** length loss is greater, so use non-expanding high-rigidity cylinders with modelling.
- Transplant and HIV-positive men (adequate CD4) have no excess infection. A lax suspensory ligament is fixed by suturing tunica to symphysis periosteum. A neophallus takes a single cylinder anchored to the pubis, with high erosion.

8 · Post-operative care

Why this matters: early care shapes the pseudocapsule, and so the final length.

- It is day surgery. Leave cylinders **partially inflated** so the pseudocapsule forms large; apply a mummy wrap, remove the drain within 24 h and keep the penis upward for 4–6 weeks. Analgesia: dorsal nerve block (liposomal bupivacaine), short opioid course, then non-steroidal anti-inflammatory drugs.
- Activation is at **4–6 weeks** with the partner present; then cycle daily. Partners are slightly less satisfied; dissatisfaction is about length and rigidity.

9 · Penile revascularisation †

Why this matters: it suits only a small, selected group.

- †Only young men (<55) with a focal post-traumatic pudendal occlusion qualify: the inferior epigastric artery is joined to the dorsal artery (or deep dorsal vein). Surgery for generalised atherosclerosis or venous leak is not recommended.

76 Diagnosis and Management of Peyronie Disease

1 · Definition

Why this matters: Peyronie disease (PD) is common and rarely resolves.

- **PD** (American Urological Association, AUA 2015): acquired penile abnormality with **fibrosis of the tunica albuginea**, with or without pain, deformity, erectile dysfunction (ED) and distress.

> **KEY POINT**
> Myths corrected: PD is **not rare** (3–20%; higher in diabetes and ED). **Spontaneous resolution is rare**: up to half worsen untreated, so "wait 6–12 months" is a myth. It affects teens to 70s, all races; flaccid trauma can trigger it; **calcification is not a marker of chronic disease**.

2 · Anatomy and pathophysiology

Why this matters: anatomy explains dorsal plaques and the direction of bend.

- Tunica: type I collagen and elastin, 1.5–3 mm, with **inner circular and outer longitudinal layers**; the septum anchors to the inner layer.
- The longitudinal layer is thinnest at 3 and 9 o'clock and **absent at 5–7 o'clock**, so the dorsum buckles; **60–70% of plaques are dorsal** with septal involvement.
- **Microtrauma**: bending during intercourse causes **delamination at the septal insertions**.
- **Transforming growth factor-β1 (TGF-β1)** is central, with other growth factors, oxidants and plasminogen activator inhibitor. Collagen becomes disorganised and elastin decreases, so the penis **curves toward the plaque**.
- A **susceptible host** is needed (penile fracture rarely causes PD); it is associated with **Dupuytren and Ledderhose** disease. Only 16–40% recall trauma.

3 · Clinical features and phases

Why this matters: the phase decides the aim.

- Features: pain, deformity (curvature, indentation, **hourglass**, hinge, shortening), plaque and ED. Hourglass or hinge gives an **unstable penis**, sometimes with distal softness.
- Kelami grading: ≤30° mild, 31–60° moderate, >60° severe. Up to 60° (especially dorsal) is often manageable; ventral and lateral are harder.

Feature	Active (acute)	Stable (chronic)
Pain	Present (flaccid, erection, intercourse)	Usually resolved
Deformity	Progressive	No progression
Aim	Prevent progression; non-steroidal anti-inflammatory drugs (NSAIDs) for pain; traction	Correct deformity (injection, surgery)
Surgery	Not appropriate	When **≥12 months from onset and stable ≥6 months**

- Differential: congenital curvature, chordee without hypospadias, epispadias, post-traumatic fibrosis and tumours (epithelioid sarcoma mimics PD).

4 · Evaluation

Why this matters: erection quality and objective curvature choose the treatment.

- **Key question**: "If straight with current rigidity, would it be adequate for penetration?" If **no**, offer prosthesis with straightening. The answer predicts post-operative erectile function.
- The **Peyronie's Disease Questionnaire (PDQ)** has 15 items in 3 domains (psychosexual, pain, bother) and no total score.
- **Examination**: palpate plaque on stretch (size unreliable). Measure **stretched penile length (SPL)** from pubis to corona dorsally at 90°; essential, as patients overestimate loss.
- **Calcification** shows on ultrasound as hyperechoic with shadowing. Extensive calcification predicts poor intralesional response and more surgery or prosthesis.

Levine calcification grade	Size	Surgery likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm, or multiple ≥1 cm	Increased

- **Intracavernosal injection (ICI) with or without penile duplex Doppler ultrasonography (PDDUS)** is required by the AUA **before any invasive treatment**. It gives goniometer curvature, indentation, hinge, calcification and peak systolic velocity, end-diastolic velocity, resistive index (RI).
- Ventral curvature is most associated with veno-occlusive dysfunction.
- Morning testosterone only if ED; biothesiometry if sensory change.

5 · Non-surgical management

Why this matters: most oral drugs fail.

- P1 Management of Peyronie's disease — *phases, evaluation, non-surgical options table, surgical algorithm*

Not in the paper answer:

- No pain and no penetration difficulty: **reassurance** (not cancer).
- Colchicine and carnitine give no benefit; Potaba shrinks plaque without straightening.
- Phosphodiesterase type 5 inhibitors (PDE5Is) treat the associated ED, not PD itself.
- Radiotherapy: no benefit, real harm.
- Most collagenase gain is in the first 3 cycles; later surgery is not made harder.
- **Oral**: AUA "should not offer" any except **NSAIDs for pain**. Vitamin E and tamoxifen give no benefit; pentoxifylline is uncertain.
- **Extracorporeal shock wave therapy (ESWT)** helps pain and function, not curvature.

Collagenase *Clostridium histolyticum* (CCH, Xiaflex)

- It is the only PD drug approved by the Food and Drug Administration (FDA, 2013). It degrades collagen **types I and III** (despite raised tissue inhibitors of metalloproteinases) and spares elastin, vessels and nerves.
- **Protocol (IMPRESS)**: 2 injections of **0.58 mg** 24–72 h apart, then **model** the penis 24–72 h after the second. Up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling is essential.
- **AUA criteria**: stable disease, curvature **>30° and <90°** and intact erections (with or without medication). IMPRESS excluded ventral plaques, extensive calcification and disease

<12 months.

- Poor response: calcification, curve ≥60°, severe indentation, hinge; better with International Index of Erectile Function (IIEF-5) ≥17.
- Bruising, swelling and pain are common; serious: **corporal rupture** (0–4.9%) and haematoma, so no intercourse until modelling ends.
- Ventral use is not routine (urethra shares the collagens).

Other intralesional agents

Agent	Mechanism	Evidence	AUA
Verapamil	Calcium channel blocker (CCB): reduces fibroblasts and extracellular matrix, increases collagenase	Randomised trials conflicting; stabilises; inferior to CCH; cheap	May offer
Interferon α-2b	Reduces fibroblasts and collagen, increases collagenase	Randomised controlled trial (RCT) vs saline: curvature reduced 27% vs 9%; flu-like symptoms (pre-treat with NSAIDs)	May administer
Nicardipine	CCB	Reduces pain and plaque; no curvature benefit	—

- Verapamil: 10 mg diluted (10–20 mL); poor if calcification, curve >90° or ventral.

Mechanical therapy

- **Penile traction therapy (PTT)** works by mechanotransduction. In the acute phase, 6–9 h/day (off 30 min every 2 h) reduces curvature, pain and surgery, downgrading grafting to plication. It gains length; needs ≥3 h/day.
- **RestoreX** (counter-bending, 30–90 min/day) improved length, curvature and IIEF-5 in an RCT.
- Vacuum is modest; the AUA makes no recommendation for traction, vacuum or combinations.

6 · Surgery

Why this matters: surgery corrects reliably but costs length or erections.

Indications

- Disease is **≥12 months from onset, with deformity stable ≥6 months**.
- Deformity or poor rigidity prevents intercourse; conservative therapy failed.

Consent

- Goal: **functionally straight** (≤20°; European Association of Urology, EAU, ≤15°). Residual/recurrent curvature: up to 16%.
- **Length loss** follows every procedure (plication more than grafting); 70–80% have already lost length, so document SPL.
- **New ED**: grafting more than plication; ~30% if pre-op rigidity is suboptimal. Pre-op erection quality predicts it best. Sensory loss ~20%, usually transient.

Choice of procedure — algorithm

ALGORITHM

1. Ask: is the erection adequate (with or without a PDE5I)?
2. If **no**, then place an inflatable prosthesis with modelling. If the curve persists, incise with or without a graft (graft if defect >2 cm).
3. If **yes**, do PDDUS. If the curve is simple and <60–70°, with no hourglass or hinge and acceptable length loss, then **plicate** (if length loss is unacceptable, graft).
4. If **yes** but the curve is complex/multiplanar >60–70°, or there is severe indentation/hourglass, hinge or a short penis (<9 cm stretched), then do **plaque incision/partial excision with graft** if the ED risk is accepted (otherwise prosthesis).

Plication (shortening)

- It shortens the **convex side opposite the plaque**. Simple, quick, least new ED; does not fix hourglass, hinge or shortening. Most common PD operation (~half).
- Nesbit failure: poor erections, shortening >2 cm and curvature >30°.

Technique	Method	Point
Nesbit (1965)	Transverse elliptical full-thickness excision opposite the curve, closed transversely	Original; for a ventral curve, lift Buck fascia and take dorsal wedges
Yachia	Vertical 0.5–1.5 cm incision closed transversely (Heineke–Mikulicz)	No tunica removed; long incisions narrow the shaft
Essed–Schroeder	Non-absorbable figure-of-eight, buried knots	First incisionless plication
16-dot	Permanent extended Lembert, 4 dots per plication	Popular; no incision
Tunica albuginea plication (TAP, Levine)	Paired partial-thickness transverse incisions (outer layer only); buried permanent suture and 3-0 polydioxanone (PDS)	Cavernosal tissue undisturbed

Plaque incision/partial excision and graft (lengthening)

- For complex deformity with **strong erections** and no veno-occlusive dysfunction; restores calibre, regains 0.5–3 cm.
- ED predictors: age >55, veno-occlusive dysfunction (RI <0.80), a large graft, ventral curve and curve >60°. Total excision is abandoned (ED).
- **Key steps**: deglove; elevate Buck fascia with the neurovascular bundle; make an H/double-Y incision or partial excision **crossing the septum**, with darted corners; sew the graft (pericardium ≤10% oversize; small intestinal submucosa, SIS, 25%) with running 4-0 PDS; plicate any residual curve.
- Pericardium and SIS are commonest; synthetics not recommended; outcome depends on selection and technique, not graft.
- Rehab: massage/stretch from 2 weeks, nightly PDE5I, traction ≥3 h/day from 3–4 weeks for 3 months.

Prosthesis with straightening

- PD with PDE5I-refractory ED: one operation; **inflatable** preferred.
- **Manual modelling** (Wilson–Delk): inflate fully, clamp the tubing and bend opposite the curve for 60–90 s, hand on shaft not glans (urethral perforation). It straightens 86–100%.

- Residual >30° or buckling: relaxing incision (cylinders deflated; cautery <30 W with Titan). Graft if defect >2 cm, never local dermis (infection).
- Residual <45°: home modelling from 4 weeks. Length loss is the commonest complaint *(see Ch 75)*.

7 · Landmark trials

Trial	Design	Result	Take-home
IMPRESS I and II	Phase III RCT, CCH vs placebo 2:1, curvature 30–90°	Curvature reduced 34% (17°) vs 18% (9.3°); bother improved	Basis of FDA approval; modelling essential
Interferon α-2b multicentre RCT	Intralesional interferon (IFN) vs saline	Curvature reduced 27% vs 9%	First RCT of intralesional therapy

★ Putting it together: andrology decisions

A · Infertile man: work-up

Why this matters: the skeleton of any infertility essay.

ALGORITHM

1. **History and examination** (both partners): toxins, testosterone use, mumps, cryptorchidism; testis volume, vasa, varicocele (see Ch 69).
2. **Two semen analyses** (abstinence 2–7 days). Low volume with acid pH suggests ejaculatory duct obstruction (EDO) or congenital bilateral absent vas (CBAVD); post-ejaculate urine detects retrograde ejaculation.
3. **Hormones** (low count or sexual dysfunction): follicle-stimulating hormone (FSH) and testosterone; if testosterone is low, luteinising hormone (LH) and prolactin.
4. **Genetics** (azoospermia, severe oligozoospermia): karyotype, Y microdeletions; CFTR (cystic fibrosis gene) in CBAVD, plus partner.

B · Azoospermia: obstructive or not

Why this matters: obstruction is reconstructable; testicular failure needs retrieval.

	Obstructive (OA)	Non-obstructive (NOA)
Testis	Normal volume	Small, soft
FSH	Normal (≤7.6 IU/L)	Raised (>7.6 IU/L)
Clues	Absent vasa, low volume, vasectomy or hernia repair	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy
Treatment	Vasovasostomy (VV) or vasoepididymostomy (VE), transurethral resection of ejaculatory ducts (TURED), or sperm retrieval for intracytoplasmic sperm injection (ICSI) (*see Ch 70*)	Microdissection testicular sperm extraction (micro-TESE) with ICSI; fix a clinical varicocele first; no retrieval if complete AZFa/b deletion

KEY POINT

† Reversal suits a younger partner wanting >1 child; ICSI suits an older partner or female factor. Copious vas fluid (even clear and sperm-free) or sperm/parts means VV; toothpaste-like fluid or none means VE (*see Ch 70*).

C · Erectile dysfunction ladder

Why this matters: erectile dysfunction (ED) is treated stepwise.

ALGORITHM

1. **Assess:** International Index of Erectile Function (IIEF-5) or SHIM score, exam, glycated haemoglobin (HbA1c), lipids, morning testosterone. ED is a cardiovascular sentinel: stratify cardiac risk (see Ch 72).
2. **Modify** risk factors and drugs; treating hypogonadism rescues phosphodiesterase type 5 inhibitor (PDE5I) failures.
3. **Shared decision:** PDE5I (never with nitrates; re-counsel before declaring failure), vacuum, intraurethral alprostadil or intracavernosal injection (alprostadil/trimix).
4. **Penile prosthesis** (inflatable three-piece) if others fail; highest satisfaction. Infected: remove all, with or without salvage (see Ch 75).
5. **Young, focal post-traumatic arterial occlusion** → revascularisation.

D · Priapism: ischaemic vs non-ischaemic

Why this matters: only ischaemic priapism is an emergency.

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Clinical	Rigid, painful	Semi-rigid, painless; after perineal trauma
Cavernous blood gas	pO_2 <30, pCO_2 >60 mm Hg, pH <7.25	Like arterial blood: pO_2 >90, pCO_2 <40
Doppler	Little or no flow	High flow, fistula or pseudoaneurysm
Treatment	Aspirate with or without irrigation, then phenylephrine (100–500 µg every 3–5 min, up to 1 h), then distal shunt with or without tunnelling, then proximal shunt. >36 h or refractory: early prosthesis	Observe; selective embolisation if persistent

KEY POINT

Sickle cell: treat as above (no waiting for transfusion); no chilled irrigation; involve haematology. Stuttering priapism: daily PDE5I, nightly pseudoephedrine, or hormonal suppression after puberty (*see Ch 73*).

E · Peyronie disease

Why this matters: phase and erections decide treatment.

ALGORITHM

1. **Active**: analgesia, traction. Collagenase (CCH): stable dorsal/lateral 30–90°, intact erections; else verapamil, interferon α-2b.
2. **Surgery**: ≥12 months from onset, stable ≥6 months; document stretched penile length (SPL) and erections (see Ch 76).
3. If **erections are good**: simple <60–70°, no hourglass/hinge, adequate length → plication; complex, hourglass/hinge or short → incision/partial excision + graft (more ED).
4. If **erections are poor** despite PDE5I → inflatable prosthesis, with or without modelling.

F · Premature ejaculation (PE)

KEY POINT

Diagnosis: latency (lifelong ≈1 min or less), inability to delay and distress, never latency alone. Treat ED first. Options: stop–start or squeeze, lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral drug outside the US), daily selective serotonin reuptake inhibitor (SSRI; paroxetine best), tramadol (off-label, addiction risk) (*see Ch 74*).